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ORIGINAL ARTICLE

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Design of a novel multi-epitope mRNA vaccine against Marburg virus using reverse vaccinology

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ABSTRACT

Aim: The Marburg virus (MARV), a zoonotic pathogen, causes a severe hemorrhagic fever for which no approved vaccine currently exists.

Methods: This study used reverse vaccinology to develop a multi-epitope vaccine against Marburg virus (MARV). To improve stability and translational efficiency, the construct was modified as an mRNA vaccine by adding a 5′ cap, 5′ UTR, Kozak sequence, tPA signal peptide, 3′ UTR, and poly(A) tail.

Results: We selected the potential 06 CTL, 06 HTL, and 04 LBL epitopes with high antigenicity, no allergenicity, and no toxicity, and the epitopes were linked with the linkers (GPGPG, AAY, and KK) and an adjuvant (*β*-defensin 2). The designed vaccine showed an antigenic score of 0.7189 with promising properties. Molecular docking reveals that the vaccine exhibits a strong binding affinity with human TLR4 (−1301.5 kcal/ mol). Meanwhile, MD simulation confirms its structural stability, and immune simulations indicate robust cellular and humoral immune responses. Ultimately, *in silico* cloning validated the capacity for effective expression in *E. coli*. Finally, secondary structure revealed that the mRNA vaccine (−418.59 kcal/ mol) was thermodynamically stable. These outcomes may contribute to the development of an experimental MARV vaccine.

Conclusion: However, further experimental evaluation is necessary to assess the efficacy of the modified mRNA vaccine for MARV prevention.

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Marburg virus; mRNA vaccine; molecular docking; MD simulation; reverse vaccinology

Introduction

A troubling increase in the development and re-emergence of viral diseases globally has been observed recently. The Marburg virus (MARV) has attracted considerable attention owing to its zoonotic origin and capacity for human-to-human transmission [1]. MARV is a negative-sense RNA virus responsible for Marburg virus disease (MVD), a highly contagious and frequently fatal hemorrhagic fever that affects both humans and non-human primates [2,3]. The virus was initially identified in August 1967 during outbreaks in Marburg and Frankfurt (Germany) and Belgrade (Yugoslavia, now Serbia), with a high mortality rate of 83–88% [4,5]. Over the years, MARV has caused multiple outbreaks, mostly in African countries, with Rwanda reporting 66 confirmed cases and 15 deaths [6]. The virus's lethal characteristics and its importance to public health make it comparable to the Ebola virus (EBOV). As a consequence, MARV is capable of transcending boundaries and causing global health crises, yet it has been recognized as a risk category 4 virus by the World Health Organization (WHO) [7]. Transmission of MARV can occur through contact with bodily fluids (urine, saliva, breast milk, amniotic fluid, and genital fluids), though transfusion and transplantation are significant routes of transmission, with detection in blood and solid organs being a common mode of disease spread [8]. Symptoms

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may appear within 2–21 days [9]. Symptoms can include fever, chills, headache, and myalgia, as well as more severe manifestations such as internal and external hemorrhaging, rash, emesis, diarrhea, shock, and multi-organ failure [10]. Due to fetal symptoms, there is no appropriate therapy or vaccine available in the market that is potentially effective against MARV. Previous vaccine alternatives have yielded varied outcomes; VEE-derived RNA replicon vaccines have demonstrated protection in animal studies but require a comprehensive safety evaluation [11]. Despite being in the experimental phase, MVA-based vaccines have demonstrated promise against the glycoprotein (GP) [12]. The significant genetic diversity of MARV, its immune evasion strategies, and the requirement for biosafety level 4 containment during development complicate traditional vaccination approaches [13]. Recent advancements in mRNA vaccine technology, including improved delivery techniques, enhanced UTR sequences, and optimized codon usage, have opened new avenues for rapid vaccine production against emerging pathogens.

The MARV genome has seven structural monocistronic proteins arranged in the following linear sequence: 3'-NP-VP35-VP40-GP-VP30-VP24-L-5', surrounded by two untranslated regions (3' and 5' UTRs) and comprising a 19.1 kb RNA genome [14]. The virus has a trimeric, highly glycosylated class I transmembrane surface glycoprotein (GP), which enables viral attachment to cells, endosomal entry, and membrane fusion [15]. VP40 (matrix protein) serves as a crucial factor in transcription/replication, assembly, release, immune suppression, and the budding of virions [16,17]. Additionally, amplifies the virus's pathogenicity by impeding interferon signaling through the suppression of Janus kinase phosphorylation, thereby aiding the virus in evading the immune system and modifying the host's tropism [18]. On the other hand, VP24 is also a small matrix protein that aids in the development of the nucleocapsid and its translocation to budding sites, potentially affecting transcription and replication [19,20]. Conversely, NP, VP35, VP30, and L coalesce to create the nucleocapsid complex, which encapsulates the viral DNA [21,22]. At the same time, the nucleoprotein NP is crucial for the assembly of the nucleocapsid and the encapsulation of viral DNA. Subsequently, VP35 is a vital polymerase cofactor that interacts with NP and L polymerase, which are necessary for viral RNA transcription and replication [21,23,24]. Additionally, VP30 is crucial for effective viral amplification and contains several phosphate groups, which likely modulate transcription [25,26]. Finally, the L protein is essential for generating viral RNA by facilitating polyadenylation and interacting with VP35 to function efficiently for viral transcription and replication [25,27]. Consequently, we design an *in silico* vaccine that integrates immune-dominant epitopes to stimulate humoral and cell-mediated immune responses. The aim of developing an mRNA multi-epitope vaccine containing T-cell and B-cell specific epitopes with a coated specific untranslated region to enhance the immune system and avert future viral infections.

Our research introduces a novel multi-epitope mRNA vaccine candidate that targets several epitopes derived from the whole proteome of MARV, utilizing sophisticated bioinformatics methods and tools for accurate design. Our computational research utilizes epitope prediction with worldwide coverage, docking with the TLR4 receptor, normal mode analysis, and molecular dynamics (MD) simulation, immune simulation, *in silico* cloning, and mRNA secondary structure prediction to enhance vaccine efficacy by thoroughly managing the immune response, emphasizing both humoral and cellular components. The primary limitation is the reliance on *in silico* validation, together with the lack of experimental *in vitro* and *in vivo* testing to assess the vaccine's true efficacy.

Materials and Methods

Sequences retrieval & characterization of proteins

We retrieved the whole protein sequence of Marburg virus (strain Musoke-80) from the UniProtKB database (https://www.uniprot.org/) with organism-specific filtering [28]. Specifically, all proteins, including glycoprotein, nucleoprotein, VP40, VP35, VP30, VP24, and L protein, were used for further immunoinformatics investigations, including antigenicity, transmembrane topology, epitope prediction, and physicochemical analysis. Subsequently, we utilized the Vaxijen v2.0 server (http://www.ddgpharmfac.net/vaxijen/) for protein antigenicity prediction, while the TMHMM v2.0 server (http://www.cbs.dtu.dk/services/TMHMM/) was employed to elucidate protein localization and membrane interaction [29,30]. Finally, the ExPASy ProtParam tool (https://web.expasy.org/protparam/) was used to assess essential physicochemical parameters, including molecular weight, solubility, stability, and hydrophobicity [31].

CTL epitope prediction and scanning

We used the NetMHCpan EL 4.1 algorithm from the IEDB web server (http://tools.iedb.org/mhci/) with the following parameters: percentile rank threshold ≤2.0 for strong binders, and comprehensive HLA class I allele coverage including HLA-A*01:01, A*02:01, A*03:01, A*24:02, B*07:02, B*08:01, B*35:01, B*44:02, C*04:01, and C*07:02 representing global population diversity from specific Marburg virus proteins [32]. Furthermore, we utilized VaxiJen v2.0 to assess antigenicity, and AllerTOP v2.1 (https://www.ddg-pharmfac.net/allertop_test/) was employed to evaluate allergenicity [33]. On the other hand, ToxinPred (https://webs.iiitd.edu.in/raghava/toxinpred/) was used to evaluate toxicity, and TMHMM v2.0 was used to predict the transmembrane region of epitopes, respectively [34]. Finally, the IEDBClass I Immunogenicity tool (http://tools.iedb.org/immunogenicity/) was used to assess the immunogenicity of the putative CTL epitopes [35].

HTL epitope prediction and scanning

We employed NetMHCIIpan 4.1 from the IEDB web server (http://tools.iedb.org/mhcii/)with full human HLA class I alleles to predict helper T-lymphocyte (HTL) epitopes [32]. Subsequently, we used VaxiJen v2.0 to assess the antigenicity, while AllerTOP v2.1 predicted the allergenicity of the epitopes. Additionally, we used the ToxinPred server to identify non-toxic properties as well as TMHMM v2.0 to verify the topological exposure of the epitopes, respectively.

On the other hand, we used IFNepitope (https://webs.iiitd.edu.in/raghava/ifnepitope/), IL4pred (https://webs.iiitd.edu.in/raghava/il4pred/), and IL10pred (https://webs.iiitd.edu.in/raghava/IL10pred/) to assess the efficacy of cytokine production. The multi-epitope vaccine design consisted only of epitopes that satisfied all immunological, safety, and cytokine criteria [36–38].

LBL epitope prediction and scanning

Prior to the T-lymphocytes, we further used the IEDB server (http://tools.iedb.org/bcell/) to predict linear B-cell epitopes from the selected virus proteome [39]. We utilized BepiPred Linear Epitope Prediction 2.0 [40], Emini Surface Accessibility Prediction [41], and Kolaskar & Tongaonkar Antigenicity techniques [42], using the default configurations. The anticipated epitopes were subsequently assessed for antigenicity utilizing VaxiJen v2.0 with a threshold of ≥ 0.4, allergenicity through AllerTOP v2.1, transmembrane helix via TMHMM v2.0, and toxicity evaluated through ToxinPred, ensuring that only the significant epitopes were selected for vaccine development. Finally, the IEDB Class I Immunogenicity tool was used to assess the immunogenicity of the putative LBL epitopes.

Population coverage

The IEDB website's population coverage analysis tool (http://tools.iedb.org/population) assessed the coverage of identified epitopes [43]. Data on epitopes for MHC-I and MHC-II were amalgamated, and the selection of MHC epitopes was executed universally throughout all geographical areas.

Multi-epitope vaccine construction

The comprehensive multi-epitope vaccine was designed by integrating predicted helper T-lymphocyte (HTL), cytotoxic T-lymphocyte (CTL), and linear B-cell (LBL) epitopes with appropriate linkers. Specifically, GPGPG was designated for HTL, AAY for CTL, and KK for LBL epitopes. Furthermore, we employed β-defensin 2 as an adjuvant and incorporated the EAAAK linker along with the PADRE sequence in the N-terminal. Finally, we used a 6-Histidine tag in the C-terminal of the constructed vaccine.

Physicochemical properties, antigenicity, allergenicity, topology, and solubility analysis of the developed vaccine template

We further analyzed the constructed vaccine by the ProtParam ExPASy server to predict physicochemical parameters, comprising molecular weight (MW), theoretical isoelectric point (pI), instability index (II), aliphatic index (AI), and grand average of hydropathicity (GRAVY). In the meantime, the VaxiJen v2.0 predicted the vaccine's antigenicity, while the AllerTOP v2.1 assessed the allergenicity of the constructed vaccine. Additionally, the vaccine's topology was examined using the TMHMM v2.0 approach. The toxicity of the vaccination peptide was determined using the ToxinPred service accordingly. Finally, the Protein-Sol server (https://protein-sol.manchester.ac.uk/) was used to determine vaccine solubility [44].

Prediction of secondary structure

The secondary structure of the peptide vaccine construct was predicted utilizing SOPMA (https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.

pl?page=/NPSA/npsa_sopma.html) and PSIPRED (http://bioinf.cs.ucl.ac.uk/psipred/), which evaluate the protein sequence to ascertain the composition and arrangement of alpha-helix, beta-sheets, coils, and turns with default settings [45,46]. These complementary methodologies provide comprehensive insights into the structural composition and stability of the vaccine candidate.

Tertiary structure prediction and evaluation

We used the trRosetta server (https://yanglab.nankai.edu.cn/trRosetta/), a deep learning algorithm to predict the tertiary structure of the constructed vaccine [47]. Subsequently, we used the GalaxyRefine server (http://galaxy.seoklab.org/refine) to enhance the original model [48]. Furthermore, we used the SAVES v6.0 server (https://saves.mbi.ucla.edu/), incorporating PROCHECK & ERRAT score and the ProSA web server (https://prosa.services.came.sbg.ac.at/prosa.php) with a range of −10 to 10, as well as evaluating structural quality, stereochemistry, overall model integrity, and sequence-structure compatibility [49–51].

B-cell conformation

Following the prediction of the vaccine's tertiary structure, we incorporated the ElliPro tool on the IEDB website (http://tools.iedb.org/ellipro/) to identify the conformational (discontinuous) B-cell epitopes in the multi-epitope vaccine based on solvent-accessibility and flexibility [52].

Molecular docking with toll-like receptors (TLRs)

Molecular docking studies were widely performed using the ClusPro web server (https://cluspro.org) (ab initio protein–protein rigid docking) to interact with human toll-like receptor (TLR4; PDB ID: 4G8A) and a multi-epitope vaccine [43]. This TLR4 was taken from the RCSB Protein Data Bank with default settings [53]. Additionally, we analyzed PDBsum (http://www.ebi.ac.uk/thornton-srv/databases/pdbsum/) to look at and describe protein–protein interactions and interface residues [54]. Finally, the optimal docking complex with the lowest energy score was visualized by Pymol software [55].

Normal mode analysis

To evaluate the structural stability and flexibility of the docked complex, a molecular dynamics (MD) simulation was performed using the iMODS server (Internal Coordinates Normal Mode Analysis Server), accessible at (https://imods.iqfr.csic.es/) [56]. The normal mode analysis (NMA) method demonstrated the multi-epitope vaccine's coordinated movement in internal coordinates. Additionally, the flexibility, B-factor, eigenvalues, variances, covariance, and elastic model were taken into account while evaluating the stability of the docked complex.

Molecular dynamics (MD) simulation

The protein–protein (TLR4-vaccine) complex stability was conducted using GROMACS version 2024.4 over 200 ns. The CHARMM36 force field of all-atom and TIP3P water model was assigned to set the system parameters [57,58]. Na^+ and Cl^- ions were added to establish electrical neutrality. Energy minimization was performed using the steepest descent algorithm with a maximum force tolerance of 1000 kJ/ mol/ nm. Temperature equilibration (NVT) was conducted for 100 ps at 300K using the V-rescale thermostat with a time constant of 0.1 ps. Pressure equilibration (NPT) was performed for 100 ps at 1 bar using the Parrinello-Rahman barostat with a time constant of 2.0 ps. Production MD was run for 200 ns with a time step of 2 fs, saving coordinates every 10 ps. Bonds involving hydrogen atoms were constrained using the LINCS algorithm [59].

Following the simulation, essential structural and dynamical metrics were calculated. These included RMSD (root mean square deviation), RMSF (root mean square fluctuation), SASA (solvent accessible surface area), Rg (radius of gyration), and H-Bond (hydrogen bond) stability. The output of each trajectory was examined in Microsoft Excel, and the figures were saved as (.png) files for the results section [60].

Immune stimulation and efficacy prediction

We used the C-ImmSim server (http://150.146.2.1/C-IMMSIM/index.php), an agent-based immune simulation program that mimics mammalian immune responses using 1000 steps and other default settings, to evaluate the immunogenic potential and efficacy of the multi-epitope vaccine design [61]. We monitored critical immunological parameters such as antibody concentrations, cytokine profiles, T- and B-cell populations, and memory cell generation. The findings indicated robust humoral and cellular immune responses, suggesting that the vaccination may elicit enduring protection against the MARV.

Codon optimization and in silico cloning

We employed the codon adaptation tool JCat (http://www.jcat.de) by maximizing the production of the

proteins in the *Escherichia coli* (*E. coli*) K12 strain through reverse translation of the constructed multiepitope vaccine with default settings [62]. Codon optimization for *E. coli* expression has two main goals: (1) facilitating recombinant protein synthesis for early in vitro immunogenicity validation tests and (2) offering an alternative expression system for protein-based vaccination if required in a prime-boost approach. Although our main emphasis is on mRNA delivery, the validation of protein expression systems enhances holistic vaccine development and allows for various immunization strategies [63]. Nonetheless, we recognize that codon preference and tRNA availability in *E. coli* could influence translation rates and the efficiency of protein folding [12]. To measure the transcription and translation efficiency, the server shows a codon adaptation index (CAI) greater than 0.8 and a GC content ranging from 50% to 60%. Finally, SnapGene software (https://www.snapgene.com/) was used for *in silico* cloning and to provide a visual representation of the recombinant vaccine [64].

mRNA vaccine construction

In this study, we further design a linear mRNA vaccine via connecting the 5′ end, which was modified with 7-methylguanosine (m7G) cap with a triphosphate to the first nucleotide (m7GpppN). For the vaccine efficiency, the human β-globin (NCBI ID: NM 000.518.5) UTR sequences were selected due to their well-characterized ability to enhance mRNA stability and translation efficiency. The 5′ UTR provides optimal ribosomal binding and scanning efficiency, while the 3′ UTR contributes to mRNA half-life extension through stabilization against 3′–5′ exonuclease activity. These UTR sequences have been successfully used in multiple clinical mRNA vaccine applications, providing a proven regulatory pathway for approval. Further, we used the Kozak sequence containing a start codon (Methionine) while additionally adding a tissue plasminogen activator (tPA) (UniProt ID: P00750) secretory signal sequence was incorporated at the 5′ end. Additionally, A poly (A) tail consisting of 120 nucleotides was added to the 3′-end of the vaccine. The design of the mRNA construct sequence ensured that the reading frames and order of the sequence were correct between the various components of the mRNA construct to optimize the expression of the vaccine and its immunogenicity tests.

mRNA vaccine secondary structure prediction

For mRNA vaccine efficiency, we used the VectorBuilder (https://en.vectorbuilder.com/) codon optimization tool to reverse-transcribe amino acid sequences to nucleotide sequences to be expressed in human cells [65]. Subsequently, the nucleotide sequence was uploaded to the DNA sequence >RNA sequence > Protein sequence server (https://biomodel.uah.es/en/lab/cybertory/analysis/trans.htm) to be transcribed into mRNA. On the other hand, RNAfold (http://rna.tbi.univie.ac.at/cgi-bin/RNAWebSuite/RNAfold.cgi) from the Vienna RNA Package 2.0 predicts the secondary structures of the mRNA constructs [66]. In this server, the minimum free energy (MFE) fraction was used to predict the thermodynamic structure of mRNA. Here, the MFE structure, centroid secondary structure, and minimum free energy were examined.

Results

Retrieval & characterization of viral proteome

The UniProtKB database provided the full protein sequences of the Marburg virus in FASTA format, encompassing glycoprotein, nucleoprotein, VP40, VP35, VP30, VP24, and L. We chose these proteins because of their essential functions in immune evasion, viral entry, and replication. However, the antigenicity, transmembrane topology, and other characteristics of the protein, including GRAVY, isoelectric point, and aliphatic index, are presented in Table S1 in Supplementary File 1.

Identification of CTL epitopes

We collected high-affinity cytotoxic T- lymphocyte (CTL) epitopes (based on >2% percentile rank) from the viral proteome using the IEDB MHC Class I binding prediction tool. However, the selection criteria were stringent, including antigenicity, allergenicity, topology, toxicity, and immunogenicity. From an initial screening of 160 possible CTL epitopes (Supplementary File 2), 6 top-ranked epitopes demonstrating strong antigenic potential and minimal sequence redundancy were selected for inclusion in the vaccine design, as shown in Table 1.

Identification of HTL epitopes

The IEDB MHC Class II binding prediction tool was further used to predict helper T-lymphocyte (HTL) epitopes (based on >10% percentile rank) with 15-mer peptides from the full proteome of

Marburg virus. Consequently, predicted epitopes were further subjected to screening for antigenicity, allergenicity, topology, toxicity, IFN-γ, IL-2, and IL-10, respectively. From the 36 anticipated HTL candidates (Supplementary File 3), 6 epitopes exhibiting optimal immunogenic characteristics and non-overlapping sequences were selected for the final vaccine formulation, as shown in Table 2.

Prediction of linear and conformational b-cell epitopes

Linear B-cell epitopes were predicted utilizing the IEDB server, following rigorous evaluation for antigenicity, allergenicity, topology, toxicity, and immunogenicity. From predicted 33 potential epitopes (Supplementary File 4) identified within the Marburg virus proteome, four epitopes exhibiting the highest immunogenic profiles were chosen for further analysis, as shown in Table 3.

In addition, the ElliPro program predicts five discontinuous conformational B-cell epitopes with significant antigenic potential, which are illustrated in Figure 1. Thus, these conformational epitopes, along with the vaccine's linear epitopes, induce significant humoral immune responses, as shown in Table S5 in Supplementary File 1.

Analysis of population coverage

We also used the IEDB population coverage server to find vaccine-specific potential epitopes with predominant human alleles for each coverage globally. The potential T-lymphocyte epitopes show average and worldwide coverage is 75.37 and 87.98%, respectively. Additionally, the lowest coverage occurred in Central America (41.36%), while the highest combined coverage was seen in Europe (92.62%), and other regions are illustrated in Figure 2.

Multi-epitope vaccine assembly

The multi-epitope vaccine was assembled by six helper T-lymphocyte (HTL) epitopes along with six cytotoxic T-lymphocyte (CTL) epitopes, and four linear B-lymphocyte (LBL) epitopes with particular linker GPGPG, AAY, and KK, respectively. To improve vaccine immunodominance, N-terminal β-defensin 2 was included as the vaccine adjuvant, while the

Table 1. Potential CTL epitopes.

Peptide	Antigenicity	Allergenicity	Topology	Toxicity	Immunogenicity
NPHAQGIAL	1.0162	Non-allergen	Outside	Non-toxin	0.10415
RPTALPPPV	0.9853	Non-allergen	Outside	Non-toxin	0.01489
TINLGVAVL	0.9902	Non-allergen	Outside	Non-toxin	0.11574
HPAVSIHPNL	0.9293	Non-allergen	Outside	Non-toxin	0.04005
RTFDAFLGVV	0.6155	Non-allergen	Outside	Non-toxin	0.2547
SPNLLGIYLT	1.0929	Non-allergen	Outside	Non-toxin	0.12462

Table 2. Potential HTL epitopes: inducer (+).

Peptide	Antigenicity	Allergenicity	Topology	Toxicity	IFN- γ	IL-2	IL-10
TAKSLELLLRVTTEE	0.7734	Non-allergen	Outside	Non-toxin	+	+	+
FILKFVLIHQGVNLV	0.6203	Non-allergen	Outside	Non-toxin	+	+	+
FSFSLKEKELNIGRT	2.3360	Non-allergen	Outside	Non-toxin	+	+	+
LHEITPVLKMGRTLE	0.6485	Non-allergen	Outside	Non-toxin	+	+	+
AFTLEAIIDISAYNE	0.5822	Non-allergen	Outside	Non-toxin	+	+	+
ASYDHFILPQSQGKG	0.6038	Non-allergen	Outside	Non-toxin	+	+	+

Table 3. Potential LBL epitopes.

Peptide	Antigenicity	Allergenicity	Topology	Toxicity	Immunogenicity
CFGILQE	1.4569	Non-allergen	Outside	Non-toxin	0.02508
FPFLDGL	0.7907	Non-allergen	Outside	Non-toxin	0.08034
VKGVPAWLPLG	1.0860	Non-allergen	Outside	Non-toxin	0.26926
DSGVALH	0.6934	Non-allergen	Outside	Non-toxin	0.0802

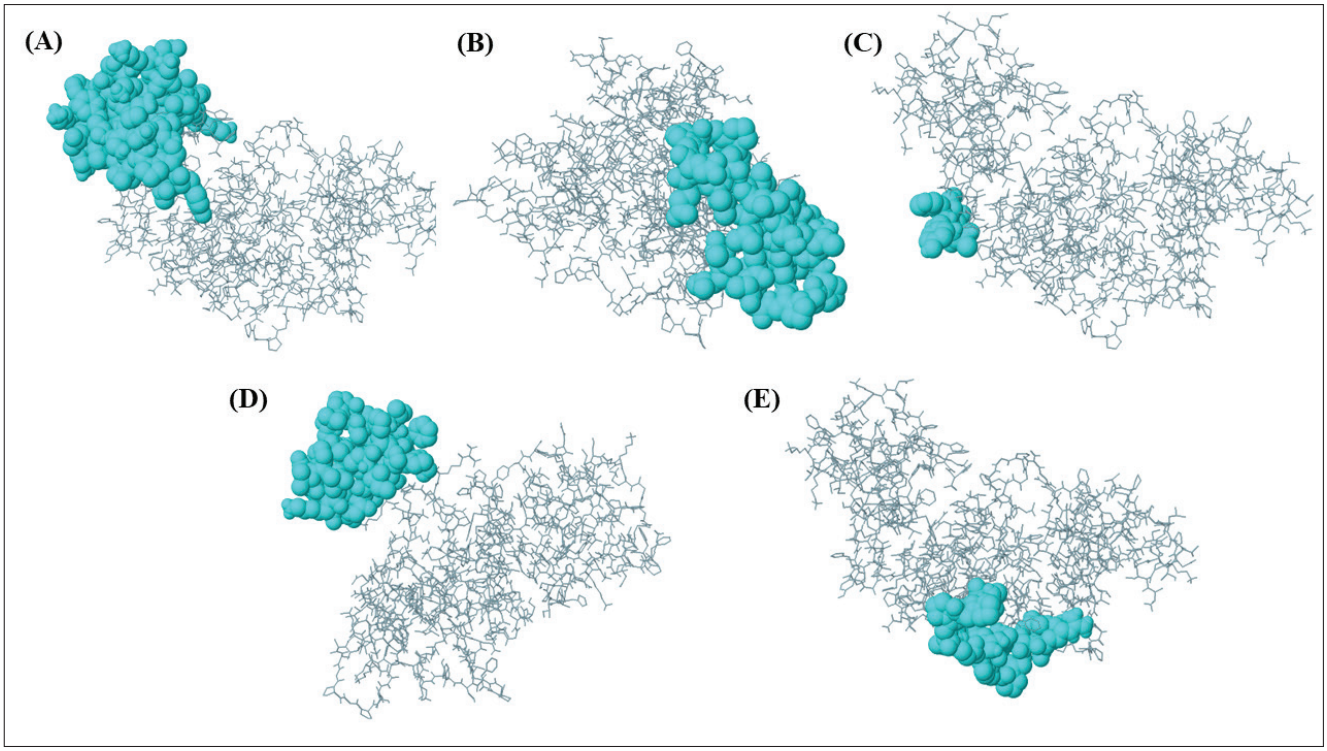


Figure 1. B-cell conformation epitopes.

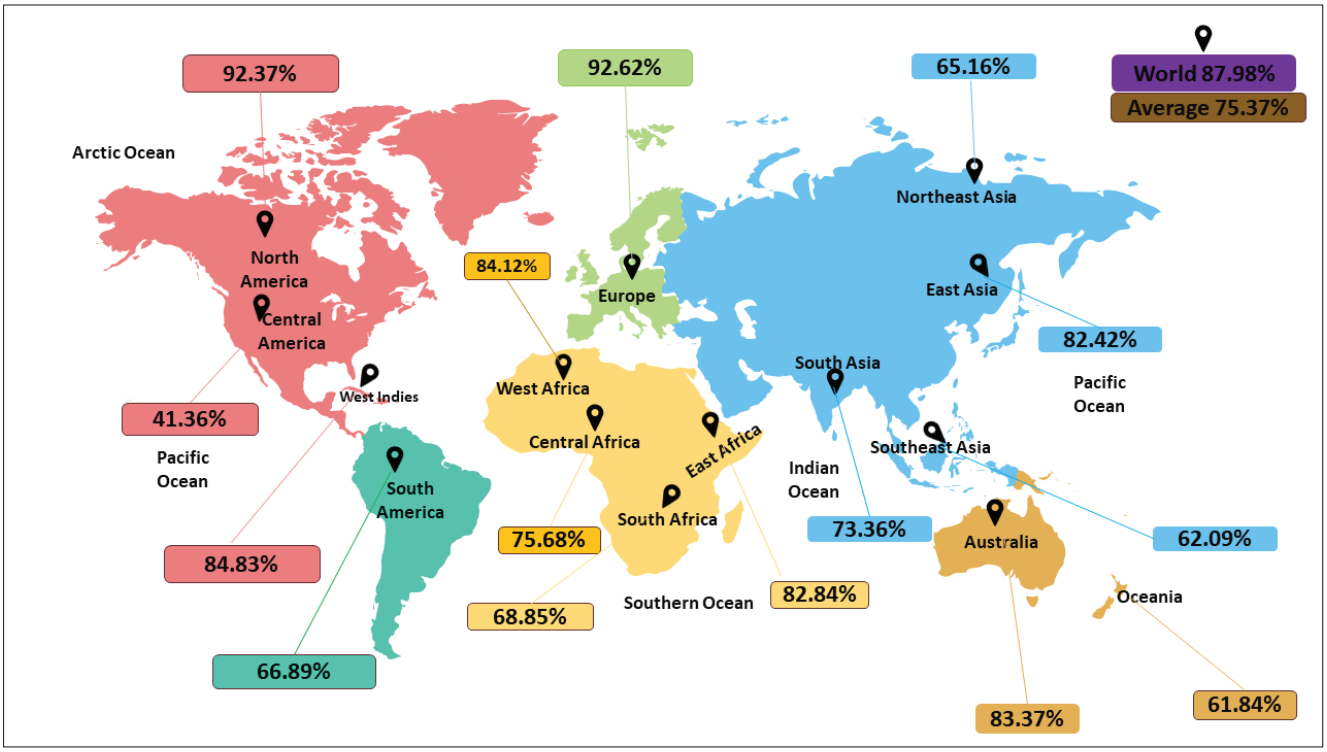


Figure 2. Population coverage of multi-epitope vaccine.

EAAAK linker with PADRE sequence was used to activate T-cell proliferation. Finally, a 6-histidine tag was used for vaccine purification in the C-terminal. The final designed multi-epitope vaccine is illustrated in Figure 3.

Vaccine analysis

Following the completion of the multi-epitope vaccine with a 300 amino acid sequence was subjected to analysis for its antigenicity, resulting in a value of 0.7189 (with threshold >0.4), suggesting a probable

antigenic. Subsequently, it exhibited non-allergenic and non-toxic characteristics, and was situated in the external transmembrane region. Furthermore, we employed the widely recognized web server Protein-Sol server to assess protein solubility, resulting in a value of 0.490, which signifies that the protein is soluble. Additionally, ProtParam analysis reveals an instability index of 32.37, indicating stability; an aliphatic index of 90.50, reflecting strong thermostability; and a GRAVY score of 0.054, which suggests modest hydrophilicity and potential solubility in aqueous solutions, as described in Table 4.

Secondary structure prediction

The peptide secondary structure was analyzed using SOPMA, revealing that the protein consists of alpha helix (23.33%), random coils (55.67%), and extended strands (21.00%). In contrast, the

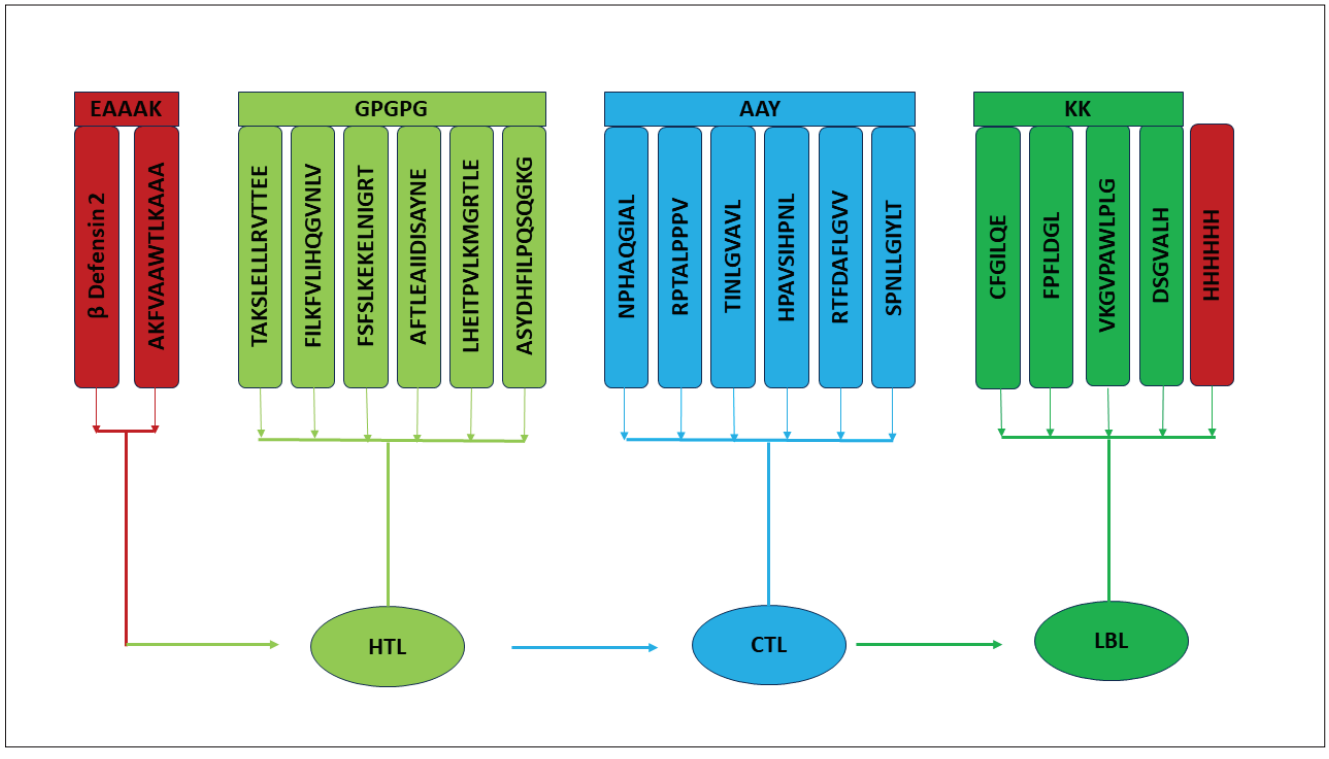


Figure 3. Multi-epitope vaccine construction.

Table 4. Antigenicity, allergenicity, toxicity, solubility, and physicochemical characteristics of the designed vaccine.

Properties	Result
Antigenicity	0.7189
Allergenicity	Non-Allergen
Toxicity	Non-Toxin
Topology	Outside
Solubility	0.490
Number of amino acids	300
Theoretical pI	9.42
Molecular weight	31,501.76
Estimated half-life	30 hours (mammalian reticulocytes, *in vitro*). >20 hours (yeast, *in vivo*). >10 hours (*Escherichia coli*, *in vivo*).
Instability index	32.37(Stable)
Aliphatic index	90.50
Grand average of hydropathicity (GRAVY)	0.054

PSIPRED web server provides graphical depictions of vaccine constructions, comprising strands (yellow), helices (pink), and coils (black). The protein's primary α-helix structures and equitable distribution of coils and β-strands suggest a stable and immunologically pertinent conformation for three-dimensional modeling and epitope mapping, as illustrated in Figure 4.

Vaccine tertiary structure modeling, refining, and analysis

The tertiary structure of the multi-epitope vaccine was generated using trRosetta, a deep-learning method for de novo protein structure prediction, to facilitate subsequent molecular recognition studies, and the top-ranked unrelaxed model was acquired. Additionally, to enhance the model's reliability, GalaxyRefine was utilized, yielding markedly improved validation metrics, including an ERRAT score of 91.273% (surpassing 75%), and with 98.3% of residues located in the most favored regions of the Ramachandran plot, while the remaining 1.7% were positioned in additional allowed regions. Furthermore, the ProSA web online z-score of the protein is −5.92 (on a scale from −10 to 10), reflecting the protein's structural quality. These enhancements confirm the successful refinement of the model and its suitability for additional structural and interaction investigation, as illustrated in Figure 5.

Molecular docking with toll-like receptor

Molecular docking was performed using ClusPro v2 to assess the vaccine's binding affinity with the human antibody-producing surface toll-like receptor to activate the immune system. At this point, the energy calculation indicated that the vaccine with TLR4 interaction has the lowest energy, −1301.5 kcal/ mol, whereas the Centered Energy is −1056.8 kcal/ mol. Additionally, the interactions at the docking interface of the vaccine-TLR4 complex are assessed by PDBsum findings, which indicate the formation of salt bridges, hydrogen bonds, and disulfide bonds between proteins, as illustrated in Figure 6.

Normal mode analysis

To perform server-based MD simulation via Normal Mode Analysis (NMA), we utilized the iMODS server to assess the TLR-4 complex conformational flexibility and dynamics. In this study, the deformability plot demonstrated that the TLR-4 complex is flexible, notably at residues 600–800 (Fig. 7A), and the B-factor comparison showed strong concordance between theoretical NMA-derived and experimental PDB-derived B-factors for both structures, supporting dynamic expectations (Fig. 7B). After eigenvalue analysis, the TLR-4 complex has shown little bit higher eigenvalue (9.386372e−05), indicating a moderately flexible and deformable structure (Fig. 7C). On the other hand, the variance plot shows higher cumulative variance in fewer modes, indicating that its inherent movements were more compact

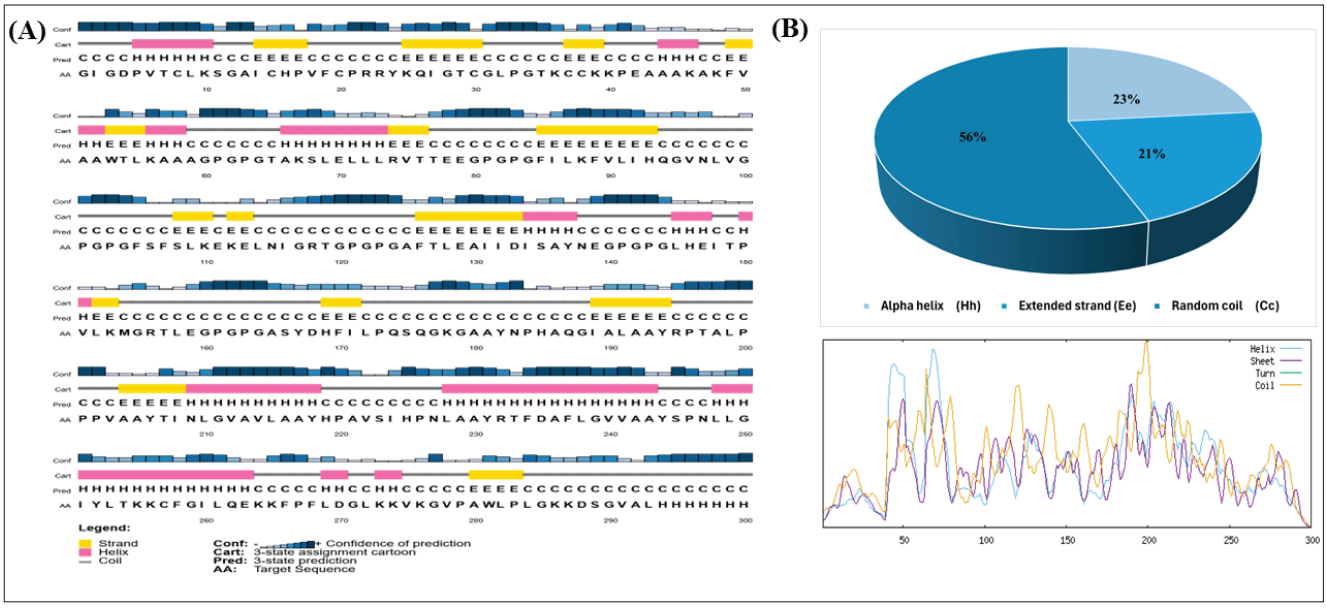


Figure 4. The secondary structure of the vaccine: (A) PSIPRED, and (B) Sopma.

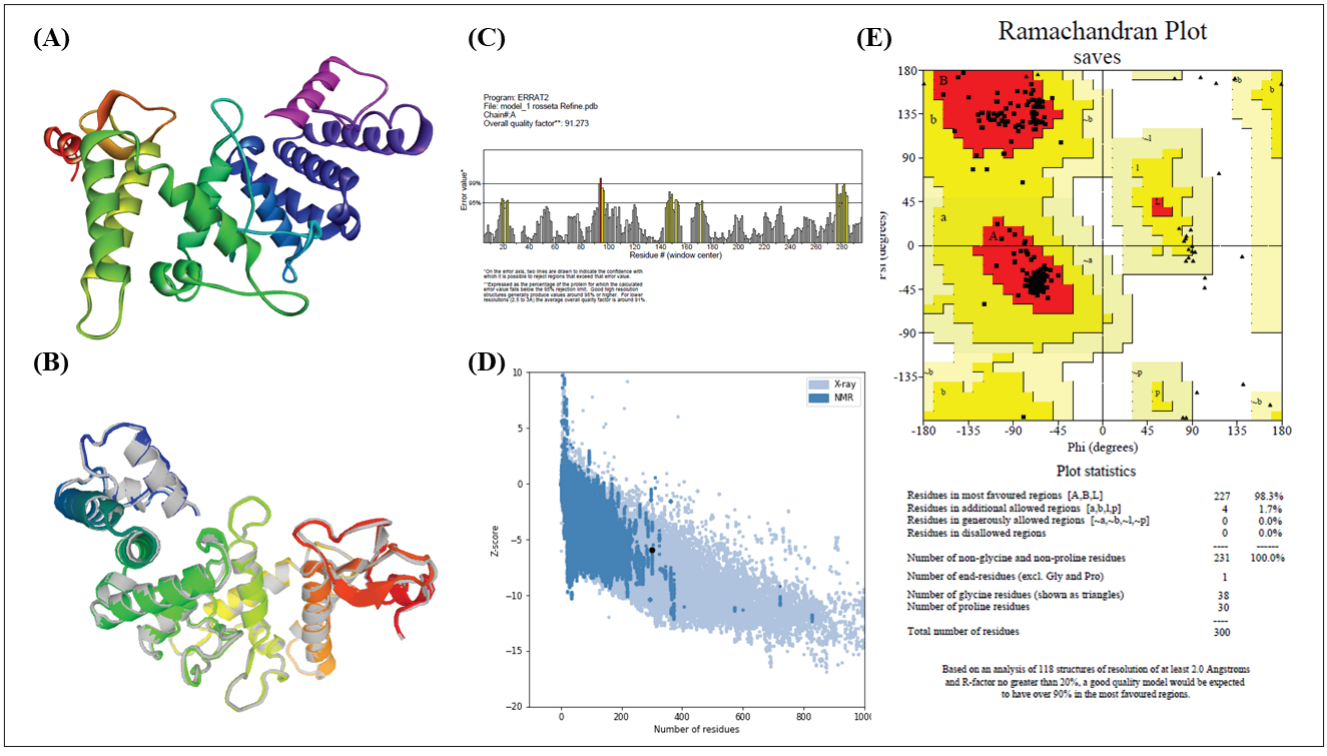


Figure 5. Predicting the tertiary (3D) structure and validating the vaccine design: (A) Tertiary Structure Model; (B) Refine tertiary structure model; (C)ERRAT quality score; (D) Z-score graph; (E) Ramachandran plot.

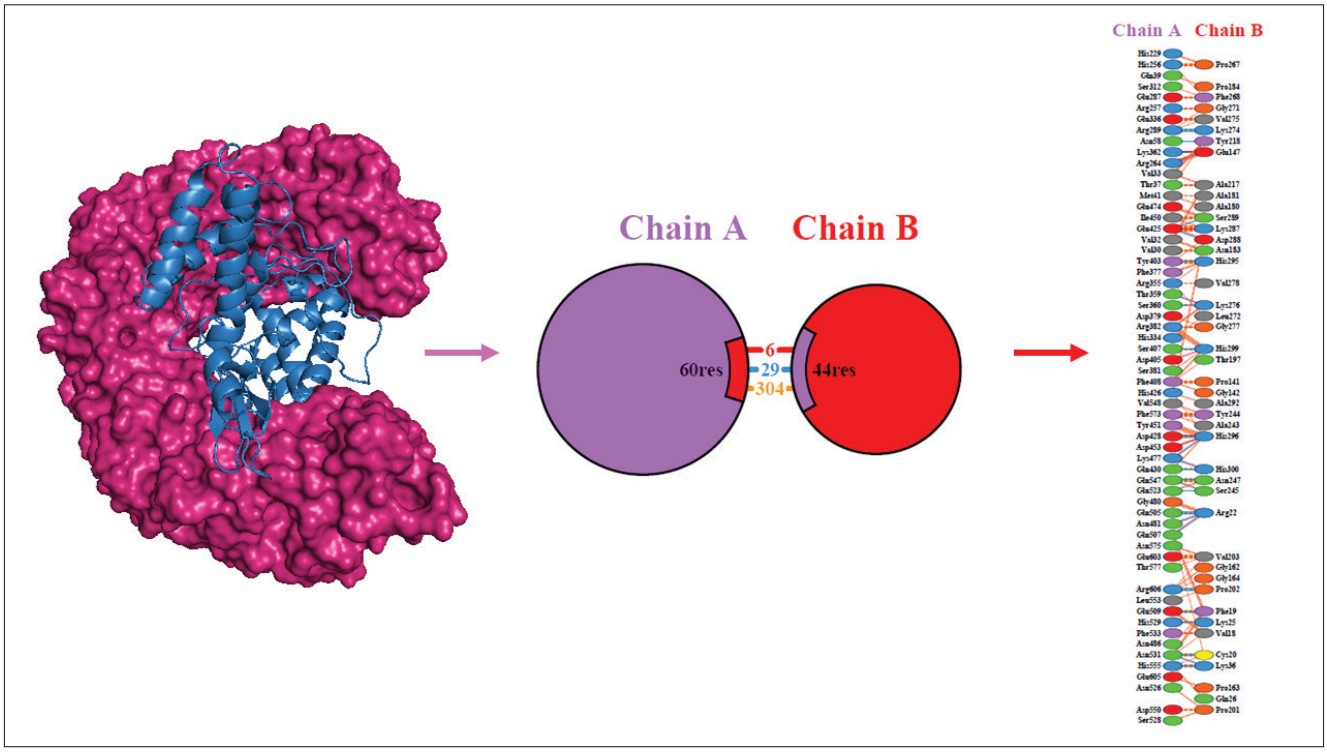


Figure 6. Docked complex of TLR4 (Hot Pink) and vaccine (Marine Blue), and protein–protein interaction of TLR4-vaccine complex.

and efficient (Fig. 7D). Additionally, the covariance matrix demonstrated that the TLR-4 complex had more correlated and anti-correlated motions, indicating stronger domain communication and coordination (Fig. 7E). Finally, the elastic network model for the TLR-4 complex showed that mechanical coupling was stronger in locations with denser and more localized atomic connections (Fig. 7F).

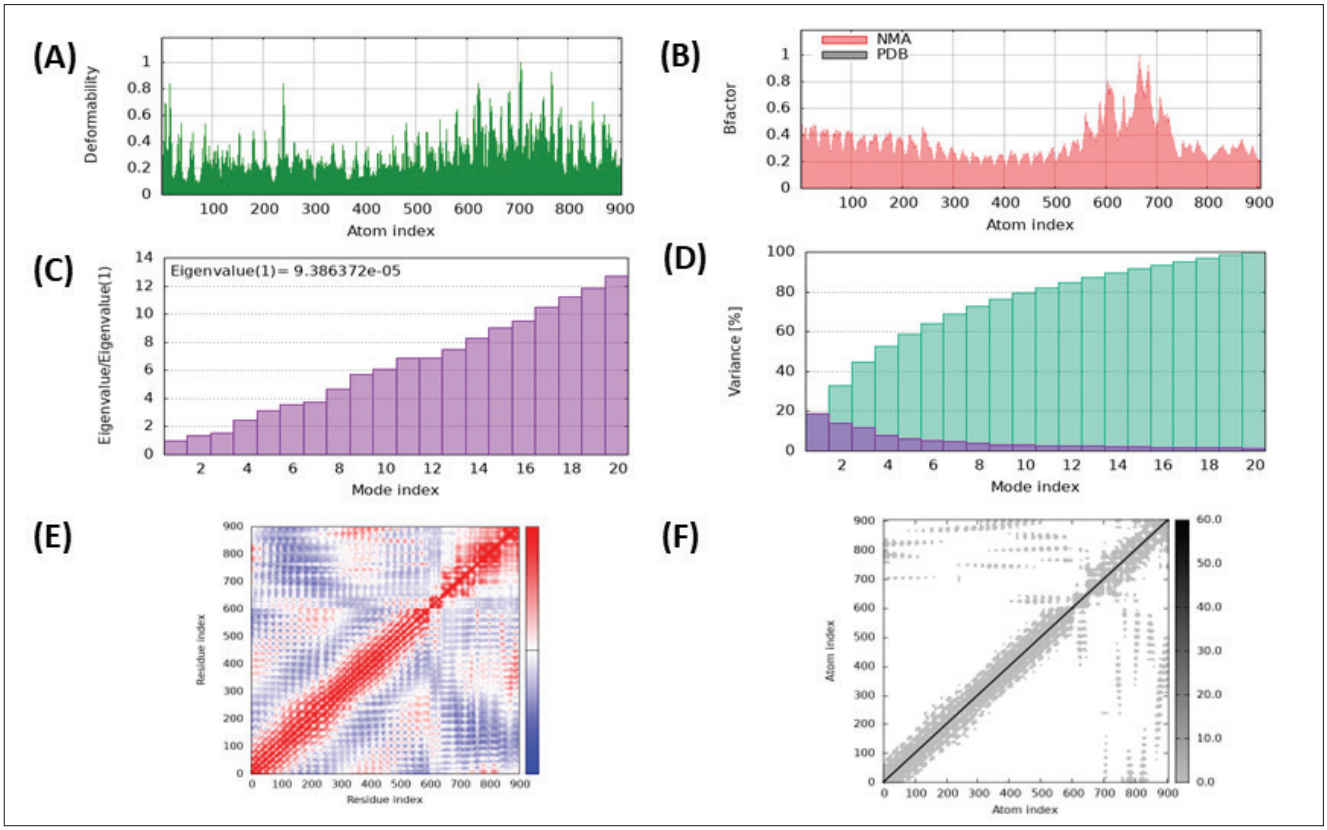


Figure 7. Normal mode analysis of the docked complex. (A) Deformability of the complex; (B) B-Factor graph, the plot indicates a comparative PDB plot, but there are no validated structures of our molecules on PDB; therefore, the plot is blank; (C) The eigenvalue plot illustrates the minimum energy required to deform the complex; (D) Variance; (E) Covariance Map correlated (red), uncorrelated (white), or anti-correlated (blue) motions (F) Elastic network (grey matter indicates stiffer region).

These computational findings show that the TLR-4 complex is dynamically adaptable and flexible, which is rigid and structurally stable (Fig. 7).

Molecular dynamics (MD) simulation

Our study utilized a 200 ns molecular dynamics (MD) simulation to study the structural stability, flexibility, compactness, solvent exposure, and intermolecular interactions of the TLR4-vaccine complex. The dynamic behavior of this complex was assessed by root mean square deviation (RMSD), root mean square fluctuation (RMSF), solvent-accessible surface area (SASA), radius of gyration (Rg), and hydrogen bonding Figure 8A–9F.

Firstly, the RMSD analysis shows that the TLR4-vaccine complex stabilized very rapidly in the early simulation and remained at a constant level over the whole 200 ns trajectory. The maximum and minimum range of RMSD values were ~0.30–0.40 nm, indicating the better stability of this complex, but with small fluctuations in between. There were also some instances of decreased RMSD values, suggesting that these were due to a change from one conformer to another rather than an unstable structure (Fig. 8A).

Secondly, most areas of the TLR4 residues, as determined by evolving time RMSF plots, showed relatively little change, with the majority being <0.5 nm. However, more significant variations were observed in some terminal/loop regions where the maximum RMSF recorded ~0.8–0.9 nm remains stable backbone (Fig. 8B). In the case of the vaccine, its maximum RMSF values were ~0.1–0.4 nm, with some regions (linker/epitope) peaking at ~0.6–0.7 nm, showing structurally stable (Fig. 8C).

Moreover, the SASA profile of the TLR4-vaccine complex has remained stable throughout the entire experiment, fluctuating between ~365 and 385 nm^2 without displaying any abrupt jumps or decreases in value. This indicates that there have been no significant unfolding or collapse events of the TLR4-vaccine re-complex, indicating the stability of the tertiary structure (Fig. 8D). In addition, the average Rg was about 0.9 to 1.0 nm, and maxima at random times during the simulation were observed, showing very low amounts of gyration;

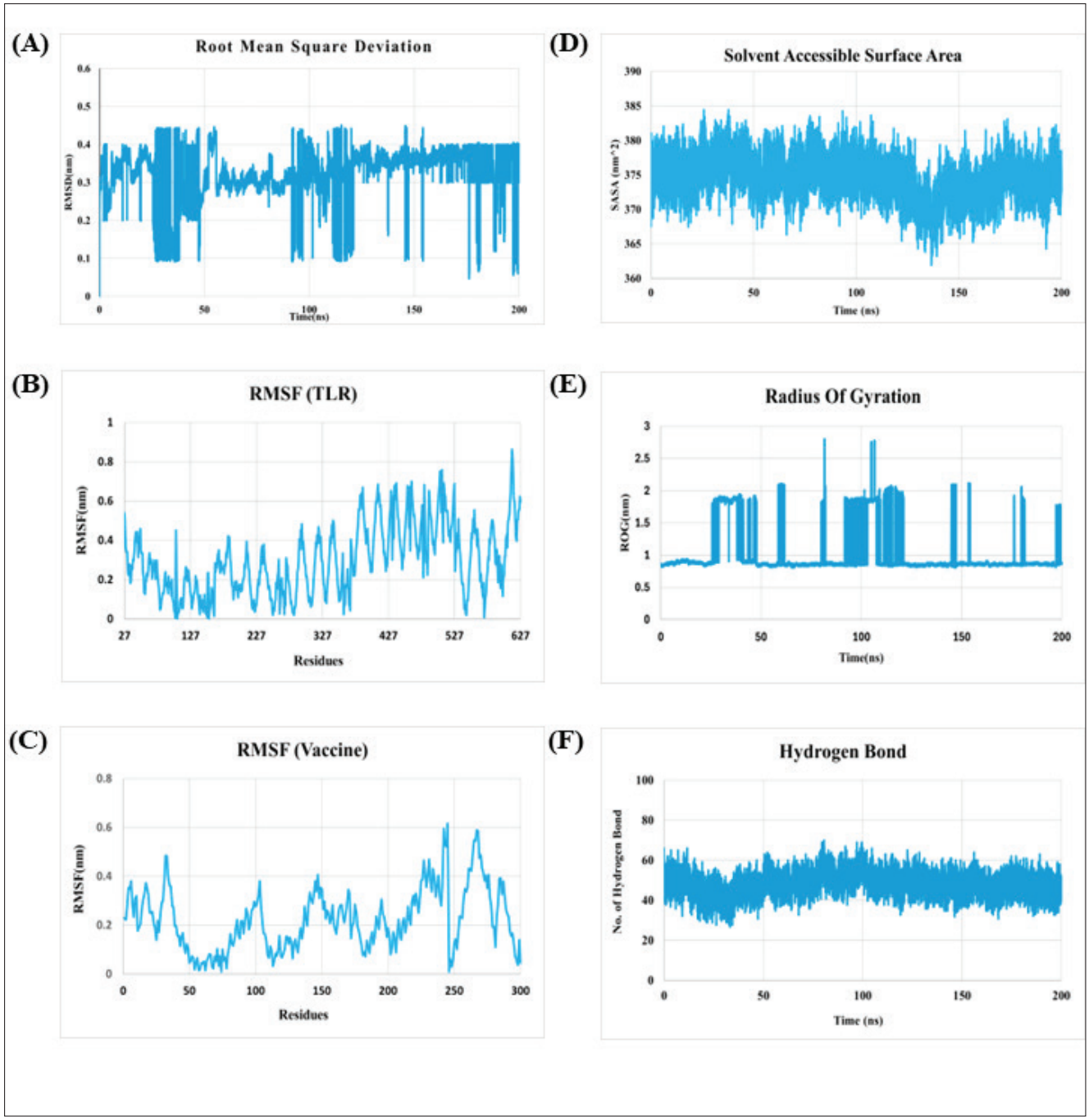


Figure 8. MD simulation of the TLR4-vaccine complex: (A) Root-Mean-Square Deviation (RMSD); (B) Root-Mean-Square Fluctuation (RMSF) of TLR4; (C) Root-Mean-Square Fluctuation (RMSF) of vaccine; (D) Radius of Gyration (Rg); (E) Solvent Accessible Surface Area (SASA); and (F) Hydrogen bond formation.

as such, the complex has a very high density in its packing (Fig. 8E).

Lastly, over the course of 200 ns of simulation time, the hydrogen bond count between TLR4 and the vaccine remained relatively stable, with no significant change (between ~40 and ~60), providing evidence of continued strong interactions (Fig. 8F). Detailed hydrogen bond persistence analysis throughout the 200ns trajectory revealed that 78% of initial vaccine-TLR4 interactions identified in PDBsum remained stable (>70% occupancy) throughout the simulation. Key binding residues, including Asp299, Asp301, and Phe303 of TLR4, maintained consistent contact with vaccine epitope regions. Specifically, the β-defensin 2 adjuvant region showed the most stable interactions with TLR4, with 5–7 persistent hydrogen bonds maintained throughout the simulation, supporting its role in sustained immune activation.

Immune simulation of the designed vaccine

Following the administration of the multi-epitope vaccine, an *in silico* immunological simulation demonstrated a strong and balanced host immune response. In this scenario, Ag levels rapidly escalated and subsequently diminished, while B cells show an innate immune response and release antibodies, specifically IgG+ IgM, IgM (Fig. 9A). Following IgM, IgG subtypes predominated, signifying a robust humoral immune response. Consequently, B cells and memory B cells proliferated, although isotype-switched B cells (IgG1 and IgG2) persisted at elevated levels, signifying sustained humoral immunity (Fig. 9B). Cytokine profiling subsequently revealed higher levels of IFN-γ, indicating a strong cellular immune response, while IL-2, IL-4, and IL-10 signified a mixed Th1/Th2 response and adequate immunoregulation (Fig. 9C). A comprehensive examination of the NK cells demonstrated stability with little change, suggesting restricted activation of the innate immune response (Fig. 9D). Furthermore, the immune system is operating effectively as CD8+ T cells shifted from replication to active and resting states, whereas cytotoxic T cells progressively augmented (Fig. 9E). Moreover, helper T cells (TH) proliferated, and TH memory cells emerged, thereby augmenting long-term adaptive immunological memory (Fig. 9F). The data indicate that the multi-epitope vaccine design is exceptionally immunogenic, eliciting robust humoral and cellular immune responses with enduring memory.

Vaccine codon optimization and in silico cloning

For vaccine cultivation in a host organism (especially *E. coli*), the Codon Adaptation Index (CAI) indicates that the modified codons of the vaccine comprise a larger proportion of the most commonly occurring codons. The increased GC content of 56.3 and a CAI value of 0.91 were utilized to identify the modified codons. The absence of SacI and BssHII restriction sites in the pET28a (+) expression vector proved that the vaccine sequence had been successfully introduced. After that, the pET28a (+) vector was implemented to produce a 4921 base pair clone. The desired region in red between each pET28a (+) vector sequence is illustrated in Figure 10.

mRNA vaccine assembly

In this study, the mRNA vaccine was developed by combining the 5′ cap for enzymatic capping, the 5 ′ UTR (human β-globin) for protein expression, the Kozak sequence to initiate the translation process, and tPA (signal peptide) for antigen secretion.

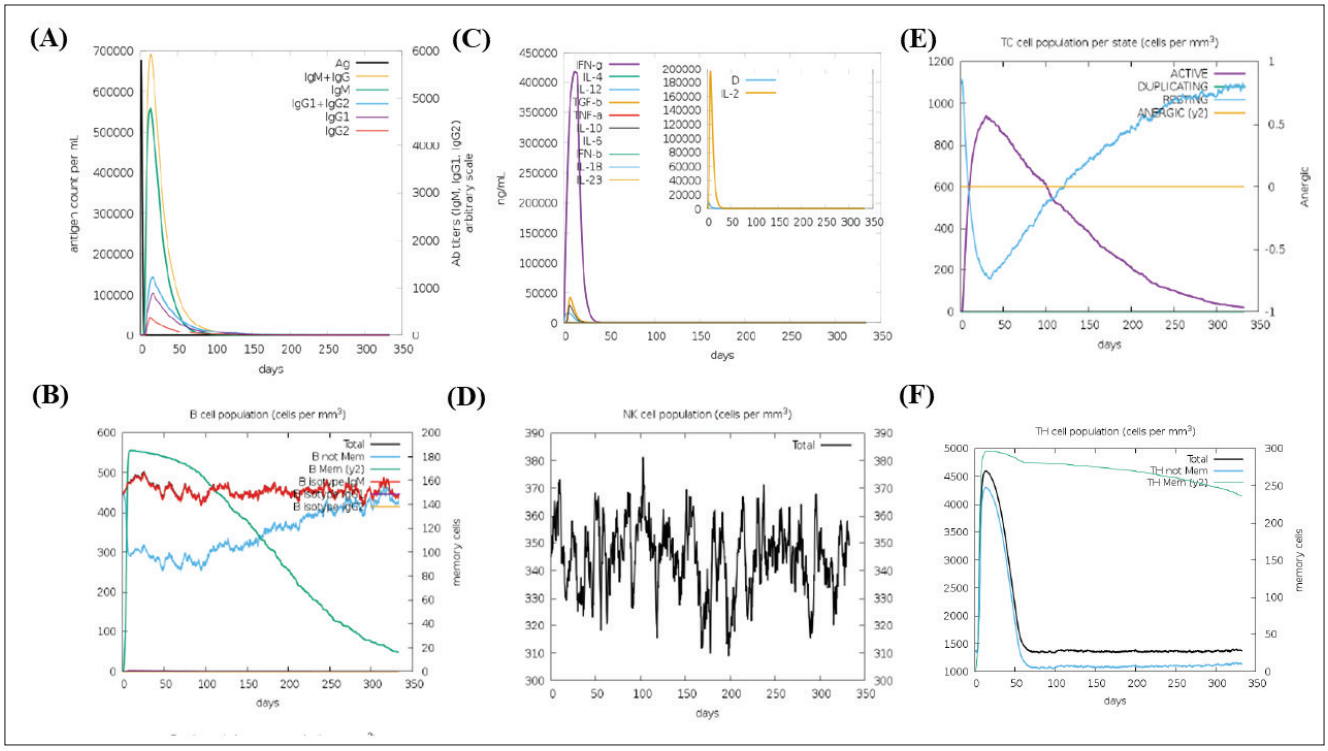


Figure 9. Immune simulation after immunization generated by C-ImmSim. (A) Antibody titer population; (B) B-cell population; (C) IL and cytokine; (D) NK cell population; (E) TC cell population; and (F) TH cell population.

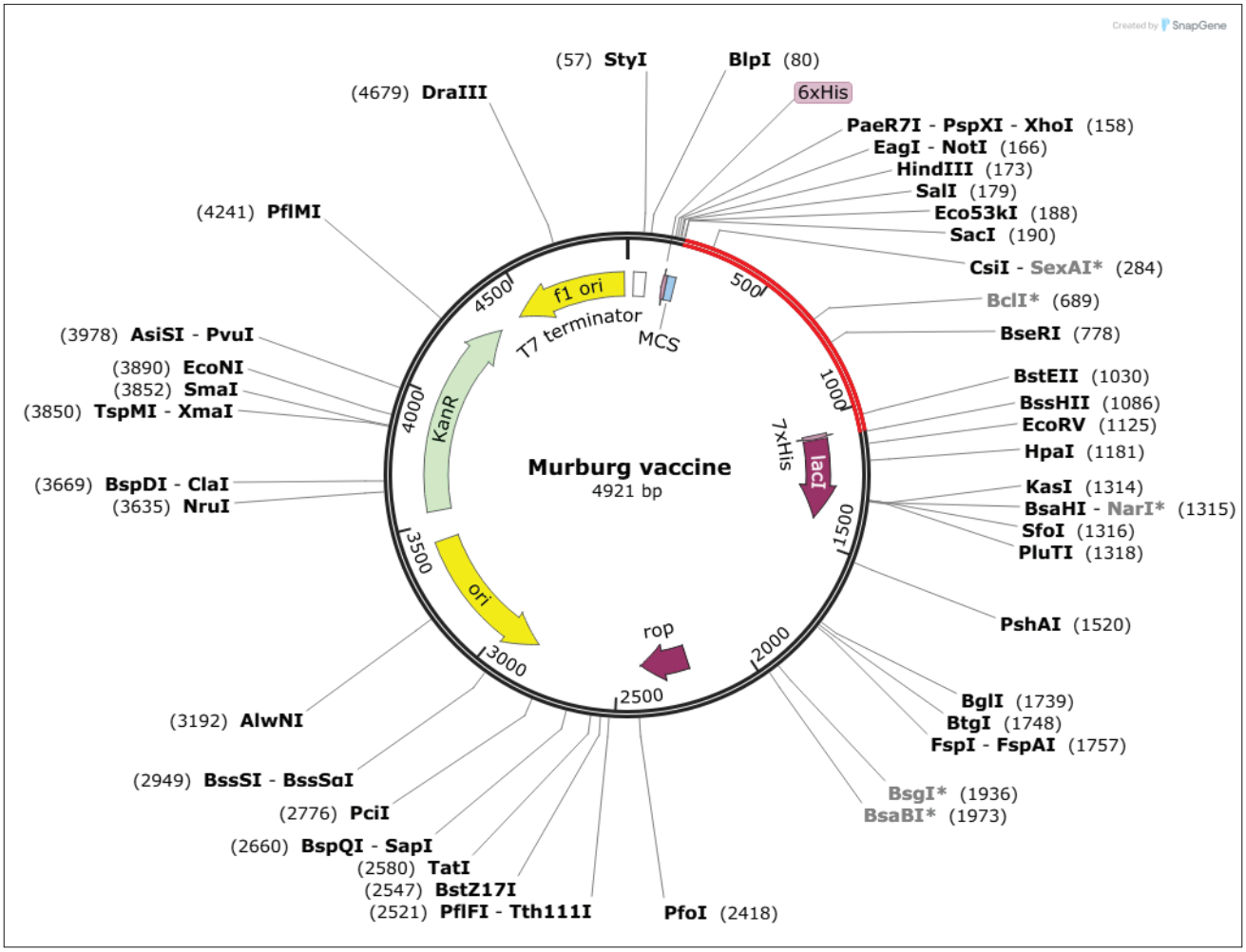


Figure 10. *In silico* cloning of a multi-epitope vaccine.

At the 3′ end of the mRNA vaccine, we integrated a poly (A) tail for nuclear export mimicry, and the 3′-UTR derived from human β globin for increasing translation duration, as illustrated in Figure 11. Finally, the constructed mRNA vaccine sequence is shown in Supplementary File 1.

Gene expression and secondary structure prediction

The designed mRNA vaccine was optimized by using the Vector Builder codon optimization tool to enhance translational efficiency in human host cells. The optimized mRNA sequence had a total length of 1120 base pairs with a Codon Adaptation Index (CAI) value of 0.90, indicating favorable expression potential in the host system. Furthermore, the optimized design shows an average GC content of 59.69% (optimal range: 30–70%) for efficient mRNA stability and expression. On the other hand, the secondary structure of mRNA was predicted by RNAfold (ViennaRNA Package) server to generate consistent 2D structures. RNAfold calculated a free energy of the thermodynamic ensemble at −418.59 kcal/ mol, indicating significant thermodynamic stability. Analysis of the predicted MFE structure specifically around the Kozak sequence region confirmed the absence of stable hairpins in the start codon vicinity. The region spanning positions showed predominantly single-stranded structure with minimal secondary structure formation, supporting efficient ribosomal binding and translation initiation. The 5′ UTR structure analysis revealed optimal accessibility of the ribosome binding site, with no competing structures that could impede translation. Subsequently, the ensemble diversity measured 245.97, reflecting a wide range of energetically similar conformations. The server displayed −336.94 kcal/ mol MFF, with a centroid secondary structure. These results suggest a significant probability of mRNA stability after synthesis. These results from RNAfold indicate that the mRNA construct shows notable thermodynamic stability and structural flexibility, essential for proper folding, protection from degradation, and effective translation in biological environments (Fig. 12A and B).

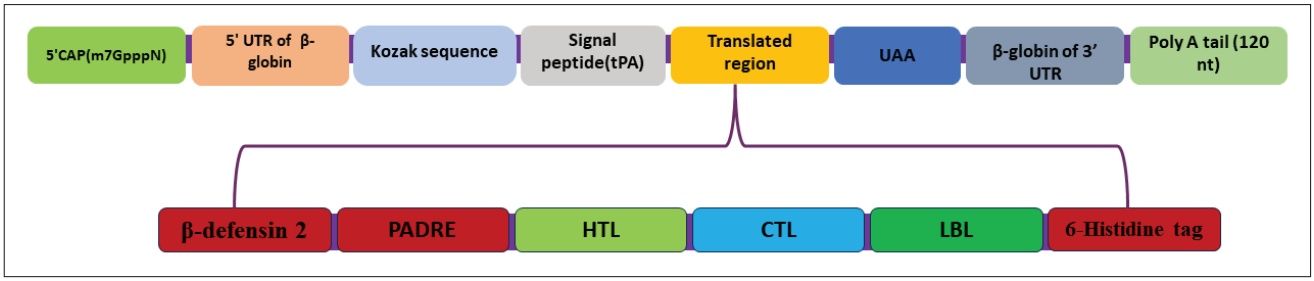


Figure 11. mRNA vaccine design.

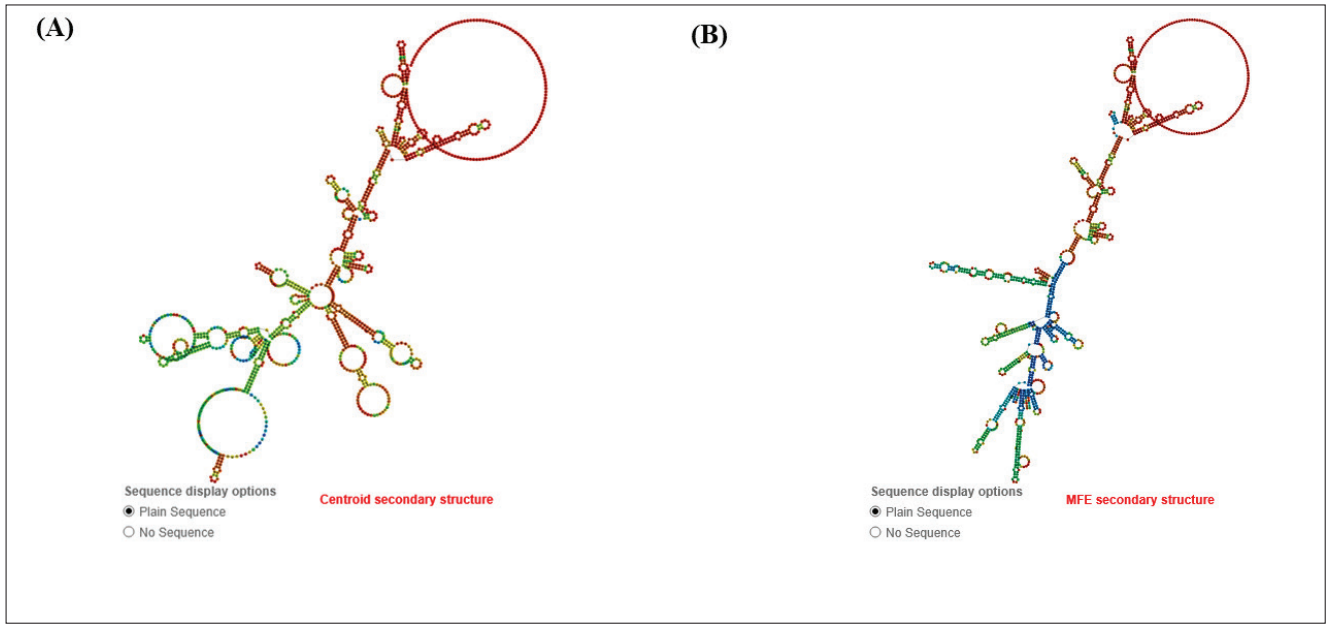


Figure 12. Secondary structure of mRNA vaccine. (A) Centroid secondary structure; (B) MFE secondary structure.

Discussion

The MARV is a fatal filovirus that causes Marburg viral disease, exhibiting zoonotic traits and the capacity to induce severe hemorrhagic fever [67]. Due to zoonotic characteristics, there is currently no MARV vaccine that has received FDA approval. To address this pressing health concern, traditional vaccines are more difficult to design for diseases with antigenic hypervariability due to compromised immune responses [68,69]. In this context, using immunoinformatics, bioinformatics, and reverse vaccinology is the way to reduce clinical trials and costly testing on living organisms to show a multi-epitope immunization strategy targeting SARS-CoV-2, HIV, Ebola, Hepatitis C, Dengue, and other pathogens to reduce *in vitro* testing and facilitate *in silico* therapy [70–72]. Comparison of our selected epitopes with known protective epitopes from related filoviruses revealed interesting homologies and differences. Several of our CTL epitopes share structural similarities with established Ebola virus T-cell epitopes, particularly in the glycoprotein and nucleoprotein regions [73]. However, our multi-epitope approach targeting the entire MARV proteome provides broader coverage than single-protein vaccines, potentially overcoming viral immune evasion mechanisms that have limited previous filovirus vaccine efficacy. Notably, mRNA vaccines have shown good results in preclinical and clinical analyses of severe diseases such as Influenza, Zika virus, and Rabies [74,75]. In this study, the MARV can be transmitted in humans as well as various bodily fluids (saliva, breastmilk, and blood) via infected bats [76,77]. This epidemiological shift emphasizes the urgent necessity for specific, safe, and rapidly deployable countermeasures. Hevey et al. demonstrate that VEE-based RNA replicon vaccines protect non-human primates and guinea pigs from MARV [78]. Furthermore, the MVA-based vaccines show protection only against the GP protein in the trial phase [12]. Additionally, multi-epitope vaccination techniques that target many viral proteins may give broader and stronger immune protection than those that target only one

antigen, such as glycoprotein (GP). Epitopes were selected from a variety of Marburg virus structural and non-structural proteins, including NP, VP35, VP40, GP, VP30, VP24, and L. Because these proteins engage in distinct stages of the viral life cycle and have immunologically important domains, their combined inclusion may boost epitope diversity and trigger both humoral and cellular immune responses [79]. On the other hand, M.N. Mashrur et al. conducted *in silico* studies to design multi-epitope-based vaccines against MARV by targeting specific proteins, while our multi-epitope vaccine design, delivered through an mRNA platform, provides a highly specific, non-replicating, and rapidly scalable vaccine candidate [5].

Consequently, Numerous bioinformatics tools were used in vaccines, several epitopes are tested for high antigenicity, allergenicity, non-toxicity, extracellular transmembrane structure, and immunogenicity with few adverse effects. The vaccine included six CTL, five HTL, and three B-cell epitopes with adjuvants (*β*-defensin2) and linker (GPGPG, AAY, KK) that promote antigen accumulation to overcome poor immunogenicity and improve *in silico* vaccine effectiveness, ensuring continuous vaccine release for an immune response [80,81]. Additionally, the vaccine construct's linker sequences were chosen based on their functional roles in the processing and presentation of epitopes. Because it accelerates proteasomal processing and enhances effective epitope presentation via the MHC class I pathway, the AAY linker was inserted between cytotoxic T-lymphocyte (CTL) epitopes. Helper T-lymphocyte (HTL) epitopes were connected by the GPGPG linker, which supported antigen processing while preserving epitope separation and minimizing junctional epitope development. Furthermore, to maintain the B-cell epitopes' separate immunogenic qualities and provide structural flexibility inside the vaccine construct, the KK linker was used between them [82].This strategic linker design maximizes immune recognition while preventing epitope interference or masking [83]. On the other hand, *β*-defensin 2 was chosen as our adjuvant over conventional lipid nanoparticle (LNP) formulations due to its dual function as both an antimicrobial peptide and TLR4 agonist, providing targeted immune activation without the need for additional adjuvant components. However, we acknowledge that optimized LNP formulations might offer superior delivery efficiency and reduced reactogenicity. While *β*-defensin 2 is naturally occurring in humans, its use as a fusion adjuvant may potentially cause localized inflammatory responses. Careful dose optimization and formulation studies will be essential to balance immunogenicity with safety. Future studies should compare the efficacy of *β*-defensin 2 with established LNP-based delivery systems to optimize the risk-benefit profile [84]. To design a successful mRNA vaccine, a 5′ cap structure (m7GpppN) to protect the mRNA, enable ribosomes to attach, and assist the body in distinguishing self from non-self [85]. Additionally, α and *β* globin were added in the 5′- and 3′-untranslated regions (UTRs) to enhance stability and translation accuracy, while the Kozak sequence with the start codon (AUG) improved translation initiation [86]. Finally, tPA signal peptides lead signal proteins into the secretory pathway while the poly (A) tail on the 3' end stops degradation and speeds up translation [87].

In this study, the designed mRNA vaccine demonstrated notable antigenic properties, attaining a prediction score of 0.7189 with other screening indicating it will not negatively impact the body for targeting the whole proteome of MARV, suggesting its capacity to bolster the immune response as well as other MARV vaccines (0.7130)[88]. The vaccine's biological properties were validated by determining a molecular weight of 300 residues, equivalent to 31.5 kDa. The instability index of 32.37 indicates robust stability, while values below 40 represent stability. The solubility study suggested that the vaccine exhibits exceptional solubility, with a solubility index of 0.490, notably surpassing the threshold of 0.45, thus confirming the superior solubility of the related proteins. The GRAVY score of 0.054 signifies both hydrophilic and hydrophobic properties, while the aliphatic index of 90.50 endorses stability at high temperatures. The vaccine consists of 55.67% random coils, 23.33% alpha helix, and 21.00% extended strands. Subsequently, the vaccine's three-dimensional structure was predicted and further refined through the Galaxy Refine server for precise functional stability [55]. Additionally, vaccine quality analysis shows a 91.273% ERRAT score, and the optimal region for the Ramachandran plot was 98.3%, indicating good structural integrity with minimal error folding, while the Z score (−5.92) indicates safety and efficiency for humans [89].

The prediction of conformational B-cell epitopes via the ElliPro server underscores the significance of accounting for the three-dimensional structure of antigens. The epitopes, specific to the three-dimensional structure of the antigen, are essential for

eliciting a robust antibody response, resulting in the identification of five conformational epitopes. Molecular docking and Normal Mode Analysis (NMA) of the vaccine and TLR4 were conducted to ensure the initiation of an immune response. Toll-like receptors (TLRs) are critical immune receptors contributing to viral inflammation [64]. The Ebola virus is closely related to MARV and can invoke immune responses by activating TLR4 [64]. The potential for receptor-mediated immune activation is indicated by the interaction between TLR4 and the desired vaccine component. TLR4 is essential for identifying viral components and initiating downstream innate immune signaling pathways, according to earlier studies. TLR4-mediated immune responses may aid in host detection and inflammatory signaling since the Marburg and Ebola viruses belong to the same viral family and share certain pathogenic characteristics. Thus, a physiologically meaningful interaction between the vaccine design and TLR4 may be implied by the observed persistent contact profile obtained from molecular docking and molecular dynamics simulations. To further demonstrate the functional significance of these receptor contacts, further residue-level interaction validation and experimental confirmation are required, even if the computational investigations show favorable binding behavior [90,91]. A high ClusPro score (more negative) indicates a stable and favorable complex. Our study found that the vaccine exhibited high negative ClusPro scores with immune receptors, TLR4 (−1301.5 kcal/ mol), indicating strong interactions [92]. Advanced machine learning consensus scoring methods have shown promise in further refining initial docking predictions in immunoinformatics, providing more robust computational frameworks for vaccine design. While our docking analysis provided strong binding affinity predictions, future studies could benefit from ensemble docking approaches that integrate multiple algorithms and scoring functions [68]. There are also salt bridges, hydrogen bonds, and other types of interactions between vaccines and TLRs using PDBsum. Furthermore, several studies revealed that the evaluation of the structural flexibility, deformability, and molecular motion of protein–protein complexes, predicting their stability and dynamics. An ideal protein–protein complex should exhibit minimal deformability, a high eigenvalue (indicating greater structural rigidity), stable mobility patterns, and strongly correlated atomic movements in covariance analysis, ensuring stability. For instance, we found that TLR4-vaccine complexes show low deformability, stable mobility patterns, and acceptable eigenvalues (9.386372e-05 for TLR4-vaccine complexes), indicating good structural stability. The covariance and elastic network models confirm well-maintained rigidity and flexibility balance, supporting vaccine-TLR binding [64].

MD simulation is used to determine the TLR4-vaccine complex's structural stability, flexibility, compactness, solvent exposure, and hydrogen bonding. The RMSD profile shows the values of ~0.30–0.40 nm, indicating that the TLR4-vaccine complex is a structurally stable complex throughout the duration of the MD simulation. Throughout the core region of the TLR4-vaccine complex, the RMSF values were relatively low (~0.8–0.9 nm), which indicates that the TLR4-bound region remained rigid and stable during the simulation. However, the observed degree of flexibility coincides with the nature of the Vaccine's multi-epitope characteristics. While some regions of the TLR4-vaccine complex must retain some degree of structural stability (~0.1–0.7 nm), other regions need to possess the ability to change their conformation, to facilitate the process of receptor interaction and recognition of the immune system. The SASA of approximately 365 nm^2 and 385 nm^2 indicates that there is very little fluctuation within the total amount of solvent relative to that of the TLR4-vaccine complex throughout the course of experimentation, which means that the TLR4-vaccine complex has maintained a constant tertiary structure without any significant unfolding or collapse of the structure. In addition, the Rg indicates that the TLR4-vaccine complex has a compact structure (between 0.9 and 1.0 nm). A high amount of intermolecular force between the TLR4-vaccine complex and the vaccine is demonstrated by the presence of an abundance (40 to 60) of hydrogen interactions. Because hydrogen bonds provide stability in a protein–protein complex, it is reasonable to conclude that their continued presence throughout the simulation indicates that the binding site has remained intact throughout all phases of the simulation, and this reinforces the thermodynamically stable nature of the complex.

In immunological simulation, immunization may elicit a robust antibody response, characterized by elevated levels of IgG and IgM. These data indicate that vaccination may elicit a robust antibody response capable of combating the disease. Concurrently, the quantity of B-lymphocytes increased post-vaccination, and similarly stimulates an augmentation of TH, NK, and TC cells, along

with heightened amounts of cellular components such as IFN-γ and memory cells. The results relate to the multi-epitope mRNA vaccine formulation and suggest that the vaccination may provoke strong immune responses in both humoral and cellular systems against MARV. To ensure translational efficacy and stability, we optimized recombinant vaccine molecules in E. coli-based expression systems with 56.33% GC content and a codon adaptation index score of 0.91, suggesting that bacteria may produce high levels of proteins. Furthermore, the two-dimensional structure of this mRNA vaccine reveals that the free energy of the thermodynamic ensemble is −418.59 kcal/ smol, indicating substantial thermodynamic stability within the manufacturing process. The findings of this study are persuasive; nonetheless, numerous limitations must be recognized.

Moreover, the mRNA platform will be investigated using advanced delivery systems such as lipid nanoparticles (LNPs) for an effective route between encouraging development *in silico* outcomes and preclinical development. Lipid nanoparticle (LNP)-based delivery systems have emerged as one of the most effective platforms for protecting mRNA molecules from degradation and facilitating cellular uptake and intracellular release. In the present study, although the designed construct demonstrated favorable immunoinformatic characteristics, the interaction between the mRNA vaccine and its delivery system was not evaluated. In addition, the incorporation of β-defensin 2 as an adjuvant sequence may enhance immune stimulation; however, excessive immune activation could potentially contribute to increased reactogenicity under certain biological conditions. Therefore, further experimental investigations are required to assess BLASTp screening against the human proteome to confirm the absence of cross-reactive epitopes & the effects of LNP encapsulation, delivery efficiency, immunogenicity, and possible inflammatory responses under in vivo conditions.

Conclusion

This study aimed to target the complete proteome of MARV to design a novel multi-epitope mRNA vaccine using an extensive reverse vaccinology approach that integrates immunoinformatics, molecular docking, and molecular dynamics simulation analysis. The designed vaccine demonstrated notable antigenic, non-allergenic, non-toxic, and favorable interactions with critical innate immune receptors, supported by strong immunological results demonstrating prolonged humoral and cellular protection. The results indicate that immunization might serve as a rapid and effective method to halt Marburg virus outbreaks. Subsequently, advanced delivery systems such as lipid nanoparticles (LNPs) are essential to confirm the effective route for safety, immunogenicity, and protective efficacy in both *in vivo* and *in vitro* areas. This project establishes a foundation for rapid response strategies to emerging filoviral and other high-risk viral illnesses that jeopardize global health by developing an innovative and prompt immunization platform.

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Conflicts of interest

The authors declare no conflict of interest.

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Author contributions

U.R.C., and A.M.: Conceptualization, project administration, methodology, software, interpretation, manuscript writing – original draft, review; M.E.U.: Methodology, formal analysis, visualization, investigation, and manuscript writing – original draft; M.J.F.J.: Methodology, formal analysis, visualization, investigation, and manuscript writing – original draft; A.H.A.: Methodology, Manuscript writing – original draft; T.R.K.R., A.H., and B.K.: Methodology, formal analysis, data curation, and investigation; S.R.T.: Supervision, project administration, investigation, manuscript writing – original draft, review, editing, and finalization.

Ethics approval

Not applicable, as this study did not involve human participants, animal subjects, or clinical data.

Declaration of competing interest

The authors declare that no financial interests, relationships, or personal links had an impact on the planning, carrying out, interpreting, or presenting of this study.

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Supplementary File 1

Table S1. Characterizations of Marburg virus proteome.

Protein Name	Antigenicity	Topology	Gravy	Aliphatic index	Instability index	Theoretical pI	Molecular weight	Number of amino acids
GP	0.5481	Outside	−0.504	71.94	45.52	5.88	74,376.19	681
NP	0.4761	Outside	0.559	85.28	52.27	4.94	77,861.86	695
VP24	0.5423	Outside	0.124	108.58	42.38	7.85	28,636.46	253
VP30	0.5636	Outside	−0.777	69.11	57.37	9.4	31,649.87	281
VP35	0.436	Outside	−0.179	90.73	44.32	9.03	36,142.77	329
VP40	0.4107	Outside	−0.285	86.57	25.46	9.4	33,793.72	303
L	0.4518	Outside	−0.227	93.03	43.1	8.41	26,7093.29	2331

Table S2. Predicted 160 possible CTL epitopes.

Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
GP	AAGLSWIPFF	HLA-B*57:01	0.9454	NON-ALLERGEN	NON- TOXIN	Outside	0.23876
		HLA-B*58:01					
		HLA-A*32:01					
	AIDFLLTRW	HLA-A*32:01	0.7623	NON-ALLERGEN	NON- TOXIN	Outside	0.16676
		HLA-B*58:01					
		HLA-B*57:01					
		HLA-B*53:01					
		HLA-A*01:01					
		HLA-A*23:01					
		HLA-B*44:02					
		HLA-A*26:01					
	AKSLELLLRV	HLA-A*02:06	0.7535	NON-ALLERGEN	NON- TOXIN	Outside	0.03228
	APIDFDPVP	HLA-B*35:01	2.4657	NON-ALLERGEN	NON- TOXIN	Outside	0.21516
	APSITTLPP	HLA-B*07:02	0.7789	NON-ALLERGEN	NON- TOXIN	Outside	0.13872
	DLAAGLSWI	HLA-A*68:02	1.3308	NON-ALLERGEN	NON- TOXIN	Outside	0.06443
	DMFPFLDGL	HLA-A*68:02	0.4571	NON-ALLERGEN	NON- TOXIN	Outside	0.16892
		HLA-A*26:01					
		HLA-B*51:01					
		HLA-A*02:06					
		HLA-A*02:01					
	ENENDCDAEL	HLA-B*40:01	0.6510	NON-ALLERGEN	NON- TOXIN	Outside	0.10749
		HLA-B*44:03					
		HLA-B*44:02					
	GAFFLYDRI	HLA-B*51:01	0.4229	NON-ALLERGEN	NON- TOXIN	Outside	0.19048
	GTKNLPILEI	HLA-A*30:01	0.7905	NON-ALLERGEN	NON- TOXIN	Outside	0.07631
	GWGLGGKWW	HLA-A*24:02	0.5747	NON-ALLERGEN	NON- TOXIN	Outside	0.01216
		HLA-A*23:01					
	HAIDFLLTRW	HLA-B*57:01	0.6653	NON-ALLERGEN	NON- TOXIN	Outside	0.22128
		HLA-B*58:01					
		HLA-B*53:01					
		HLA-A*32:01					
		HLA-A*26:01					
		HLA-A*68:01					

Continued

Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-B*35:01					
		HLA-A*23:01					
	HAQGIALHL	HLA-B*53:01	0.5231	NON-ALLERGEN	NON- TOXIN	Outside	0.17247
		HLA-B*51:01					
		HLA-A*68:02					
		HLA-B*35:01					
		HLA-B*58:01					
		HLA-A*02:06					
		HLA-B*07:02					
		HLA-A*32:01					
		HLA-B*57:01					
		HLA-B*08:01					
		HLA-A*26:01					
	KSLELLLRV	HLA-A*02:06	0.8474	NON-ALLERGEN	NON- TOXIN	Outside	0.09679
		HLA-A*02:01					
		HLA-B*58:01					
		HLA-A*32:01					
		HLA-B*57:01					
		HLA-A*02:03					
		HLA-A*30:01					
		HLA-A*68:02					
		HLA-A*30:02					
		HLA-B*51:01					
	LAAGLSWIPF	HLA-B*58:01	1.4437	NON-ALLERGEN	NON- TOXIN	Outside	0.18925
		HLA-B*57:01					
		HLA-B*35:01					
		HLA-B*53:01					
	LELLLRVTT	HLA-B*40:01	0.7661	NON-ALLERGEN	NON- TOXIN	Outside	0.08068
	LSIAVLIAL	HLA-B*58:01	0.8765	NON-ALLERGEN	NON- TOXIN	Outside	0.24751
		HLA-A*68:02					
		HLA-B*57:01					
		HLA-A*02:06					
		HLA-A*32:01					
	NAPIDFDPV	HLA-B*51:01	1.8389	NON-ALLERGEN	NON- TOXIN	Outside	0.27436
		HLA-A*68:02					
	NENDCDAEL	HLA-B*40:01	0.4489	NON-ALLERGEN	NON- TOXIN	Outside	0.08012
		HLA-B*44:03					
		HLA-B*44:02					
	NPHAQGIAL	HLA-B*07:02	1.0162	NON-ALLERGEN	NON- TOXIN	Outside	0.10415
		HLA-B*35:01					
		HLA-B*08:01					
		HLA-B*53:01					
		HLA-B*51:01					
	QEDDLAAGL	HLA-B*40:01	1.2182	NON-ALLERGEN	NON- TOXIN	Outside	0.10837
		HLA-B*44:03					
		HLA-B*44:02					
	SMPPHNTTT	HLA-A*02:03	0.7719	NON-ALLERGEN	NON- TOXIN	Outside	0.06609

Continued

Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-A*02:06					
		HLA-A*02:01					
	TGCFGALQEY	HLA-A*30:02	0.6181	NON-ALLERGEN	NON- TOXIN	Outside	0.1217
		HLA-B*35:01					
		HLA-B*15:01					
		HLA-A*01:01					
		HLA-A*26:01					
	TGWGLGGKW	HLA-B*58:01	1.4104	NON-ALLERGEN	NON- TOXIN	Outside	0.0297
		HLA-B*57:01					
		HLA-B*53:01					
		HLA-A*32:01					
	TGWGLGGKWW	HLA-B*57:01	0.7937	NON-ALLERGEN	NON- TOXIN	Outside	0.10752
		HLA-B*58:01					
	TTCFLISLI	HLA-A*68:02	0.9007	NON-ALLERGEN	NON- TOXIN	Outside	0.06868
		HLA-B*57:01					
	TTNEHFTSP	HLA-A*68:02	0.5345	NON-ALLERGEN	NON- TOXIN	Outside	0.17645
NP	VQEDDLAAGL	HLA-B*40:01	0.9956	NON-ALLERGEN	NON- TOXIN	Outside	0.15527
		HLA-A*02:06					
		HLA-B*44:03					
		HLA-B*44:02					
	ALHPLVRTS	HLA-A*02:03	0.6267	NON-ALLERGEN	NON- TOXIN	Outside	0.09376
	ALPPPVDNK	HLA-A*11:01	0.8092	NON-ALLERGEN	NON- TOXIN	Outside	0.02824
		HLA-A*03:01					
		HLA-A*30:01					
		HLA-A*68:01					
		HLA-A*30:02					
	APFARVLNL	HLA-B*07:02	0.613	NON-ALLERGEN	NON- TOXIN	Outside	0.15349
		HLA-B*08:01					
		HLA-B*51:01					
		HLA-B*53:01					
		HLA-B*35:01					
		HLA-A*26:01					
	APFARVLNLS	HLA-B*07:02	0.8817	NON-ALLERGEN	NON- TOXIN	Outside	0.14559
	DLLEGGLLTL	HLA-A*02:01	0.4622	NON-ALLERGEN	NON- TOXIN	Outside	0.16603
		HLA-A*02:06					
	DLNDPFALL	HLA-A*68:02	0.6587	NON-ALLERGEN	NON- TOXIN	Outside	0.14616
		HLA-A*26:01					
		HLA-B*08:01					
		HLA-A*02:03					
		HLA-A*02:01					
		HLA-B*51:01					
		HLA-A*02:06					
	DPFGSIGDV	HLA-B*51:01	0.6599	NON-ALLERGEN	NON- TOXIN	Outside	0.07784
	DSGVALHPL	HLA-B*51:01	0.7361	NON-ALLERGEN	NON- TOXIN	Outside	0.10102
		HLA-A*68:02					
	DSGVALHPLV	HLA-A*68:02	0.4703	NON-ALLERGEN	NON- TOXIN	Outside	0.09445
	GINNLEHGL	HLA-A*02:01	1.0209	NON-ALLERGEN	NON- TOXIN	Outside	0.12194

Continued

Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-A*02:06					
		HLA-A*02:03					
	GINNLEHGLY	HLA-A*30:02	0.9287	NON-ALLERGEN	NON- TOXIN	Outside	0.13066
		HLA-A*01:01					
		HLA-B*15:01					
		HLA-A*26:01					
	GLLIVKTVL	HLA-A*02:01	0.8238	NON-ALLERGEN	NON- TOXIN	Outside	0.0244
		HLA-A*02:03					
		HLA-B*08:01					
	HSLLELGTK	HLA-A*11:01	1.0804	NON-ALLERGEN	NON- TOXIN	Outside	0.12358
		HLA-A*30:01					
		HLA-A*03:01					
		HLA-A*68:01					
	HYSPLILALK	HLA-A*33:01	1.7899	NON-ALLERGEN	NON- TOXIN	Outside	0.07004
		HLA-A*30:01					
		HLA-A*31:01					
		HLA-A*68:01					
		HLA-A*11:01					
		HLA-A*24:02					
	KELQADWPDM	HLA-B*40:01	0.9623	NON-ALLERGEN	NON- TOXIN	Outside	0.15165
	LLIVKTVLEF	HLA-B*15:01	0.7211	NON-ALLERGEN	NON- TOXIN	Outside	0.00054
		HLA-A*32:01					
	LQADWPDMSF	HLA-B*15:01	1.5044	NON-ALLERGEN	NON- TOXIN	Outside	0.01594
		HLA-A*32:01					
		HLA-A*23:01					
		HLA-B*53:01					
		HLA-B*35:01					
		HLA-A*24:02					
		HLA-B*58:01					
		HLA-A*01:01					
		HLA-A*02:06					
		HLA-A*30:02					
	NSGIDLGDLL	HLA-A*68:02	1.1272	NON-ALLERGEN	NON- TOXIN	Outside	0.19986
	QTRFSGLLIV	HLA-A*68:02	0.5873	NON-ALLERGEN	NON- TOXIN	Outside	0.06446
		HLA-A*02:06					
		HLA-A*30:01					
	RFSGLLIVK	HLA-A*30:01	0.9051	NON-ALLERGEN	NON- TOXIN	Outside	0.0956
		HLA-A*31:01					
		HLA-A*03:01					
		HLA-A*11:01					
	RGRIGLFLS	HLA-A*30:01	0.791	NON-ALLERGEN	NON- TOXIN	Outside	0.2656
	RGRIGLFLSF	HLA-A*32:01	0.9359	NON-ALLERGEN	NON- TOXIN	Outside	0.1771
		HLA-B*57:01					
		HLA-A*23:01					
		HLA-A*30:02					
		HLA-B*58:01					
	RPTALPPPV	HLA-B*07:02	0.9853	NON-ALLERGEN	NON- TOXIN	Outside	0.01489

Continued

Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-B*51:01					
		HLA-B*53:01					
		HLA-B*35:01					
	RSSFILKFV	HLA-A*30:01	0.7604	NON-ALLERGEN	NON- TOXIN	Outside	0.06966
		HLA-B*57:01					
		HLA-B*58:01					
		HLA-A*68:02					
		HLA-A*32:01					
	RSSFILKFVL	HLA-B*57:01	0.5488	NON-ALLERGEN	NON- TOXIN	Outside	0.10282
		HLA-B*58:01					
		HLA-A*32:01					
	SGVALHPLV	HLA-B*51:01	0.5222	NON-ALLERGEN	NON- TOXIN	Outside	0.05658
	SLLELGTKP	HLA-A*02:01	1.3937	NON-ALLERGEN	NON- TOXIN	Outside	0.02501
	SPLILALKT	HLA-B*51:01	1.3084	NON-ALLERGEN	NON- TOXIN	Outside	0.02099
	SSFVDLNDPF	HLA-B*58:01	0.552	NON-ALLERGEN	NON- TOXIN	Outside	0.09649
		HLA-B*57:01					
		HLA-A*23:01					
		HLA-A*24:02					
	TALPPPVDNK	HLA-A*11:01	0.7942	NON-ALLERGEN	NON- TOXIN	Outside	0.01744
		HLA-A*03:01					
		HLA-A*68:01					
		HLA-A*30:01					
	YAPFARVLNL	HLA-B*08:01	0.5907	NON-ALLERGEN	NON- TOXIN	Outside	0.22842
		HLA-B*51:01					
		HLA-B*07:02					
		HLA-B*53:01					
		HLA-B*35:01					
L	YSPLILALK	HLA-A*68:01	1.5898	NON-ALLERGEN	NON- TOXIN	Outside	0.13094
		HLA-A*11:01					
		HLA-A*03:01					
	DYQENEAEL	HLA-A*24:02	0.7567	NON-ALLERGEN	NON- TOXIN	Outside	0.24262
		HLA-A*23:01					
		HLA-B*08:01					
		HLA-A*24:02					
		HLA-A*23:01					
		HLA-B*08:01					
	EDKIGYPPL	HLA-B*08:01	0.9272	NON-ALLERGEN	NON- TOXIN	Outside	0.0776
	ELFYILISKK	HLA-A*68:01	0.8915	NON-ALLERGEN	NON- TOXIN	Outside	0.01274
		HLA-A*33:01					
		HLA-A*03:01					
	GLSFLNPEK	HLA-A*03:01	1.2577	NON-ALLERGEN	NON- TOXIN	Outside	0.09635
		HLA-A*11:01					
	GSNDTELNY	HLA-A*30:02	1.1545	NON-ALLERGEN	NON- TOXIN	Outside	0.13913
		HLA-A*01:01					
		HLA-B*58:01					
		HLA-B*15:01					
		HLA-A*11:01					

Continued

Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-A*26:01					
		HLA-B*35:01					
		HLA-B*57:01					
		HLA-A*03:01					
		HLA-A*32:01					
	GSNDTELNYV	HLA-A*01:01	1.1716	NON-ALLERGEN	NON- TOXIN	Outside	0.13746
	GVAVLDIAL	HLA-A*02:06	1.2442	NON-ALLERGEN	NON- TOXIN	Outside	0.1995
	GYILFFPFGL	HLA-A*23:01	0.9972	NON-ALLERGEN	NON- TOXIN	Outside	0.3682
		HLA-A*24:02					
	HDLHLTPYL	HLA-B*40:01	1.5983	NON-ALLERGEN	NON- TOXIN	Outside	0.04317
	HDLHLTPYLR	HLA-A*33:01	1.7060	NON-ALLERGEN	NON- TOXIN	Outside	0.03591
		HLA-A*31:01					
	HLGFPLGFNI	HLA-A*02:01	1.3465	NON-ALLERGEN	NON- TOXIN	Outside	0.23412
		HLA-A*02:03					
		HLA-A*32:01					
	HYHLGGIEGL	HLA-A*24:02	0.9709	NON-ALLERGEN	NON- TOXIN	Outside	0.29492
		HLA-A*23:01					
	IGYILFFPF	HLA-A*23:01	0.9160	NON-ALLERGEN	NON- TOXIN	Outside	0.32444
	KENPYYWGML	HLA-B*40:01	1.3316	NON-ALLERGEN	NON- TOXIN	Outside	0.11405
		HLA-B*44:03					
		HLA-B*44:02					
	LFDWMHFLI	HLA-A*24:02	0.9236	NON-ALLERGEN	NON- TOXIN	Outside	0.18186
		HLA-A*23:01					
	LGFPLGFNI	HLA-B*51:01	1.0743	NON-ALLERGEN	NON- TOXIN	Outside	0.14296
		HLA-B*58:01					
	LGLDHVFSA	HLA-A*02:06	0.7566	NON-ALLERGEN	NON- TOXIN	Outside	0.09122
	LSHRSLRIL	HLA-B*57:01	1.1940	NON-ALLERGEN	NON- TOXIN	Outside	0.01248
		HLA-B*58:01					
	LSIFIKDRL	HLA-B*58:01	0.8050	NON-ALLERGEN	NON- TOXIN	Outside	0.13656
		HLA-B*57:01					
		HLA-A*68:02					
	NLNLTEYTL	HLA-B*08:01	1.2846	NON-ALLERGEN	NON- TOXIN	Outside	0.13835
		HLA-A*02:01					
		HLA-A*02:06					
		HLA-A*02:03					
		HLA-A*02:01					
		HLA-A*02:06					
		HLA-A*02:03					
		HLA-A*68:02					
		HLA-A*32:01					
		HLA-A*11:01					
		HLA-A*68:01					
	RTEILDTSPL	HLA-B*40:01	1.0168	NON-ALLERGEN	NON- TOXIN	Outside	0.06766
	SISLDPGFR	HLA-A*68:01	1.8244	NON-ALLERGEN	NON- TOXIN	Outside	0.0433
		HLA-A*11:01					
		HLA-A*31:01					
		HLA-A*33:01					

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Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
	SPKENPYYW	HLA-B*53:01	0.7902	NON-ALLERGEN	NON- TOXIN	Outside	0.00873
		HLA-B*35:01					
		HLA-B*58:01					
		HLA-B*07:02					
		HLA-B*57:01					
		HLA-B*51:01					
		HLA-B*44:02					
		HLA-B*08:01					
		HLA-B*44:03					
		HLA-A*32:01					
		HLA-A*26:01					
		HLA-A*24:02					
		HLA-A*23:01					
	SPKENPYYWG	HLA-B*53:01	1.0013	NON-ALLERGEN	NON- TOXIN	Outside	0.13647
	SQCEDFSPK	HLA-A*11:01	1.3609	NON-ALLERGEN	NON- TOXIN	Outside	0.06895
		HLA-A*03:01					
		HLA-A*30:01					
	THDLHLTPY	HLA-A*30:02	1.6292	NON-ALLERGEN	NON- TOXIN	Outside	0.04338
		HLA-A*01:01					
		HLA-B*35:01					
	TINLGVAVL	HLA-B*08:01	0.9902	NON-ALLERGEN	NON- TOXIN	Outside	0.11574
		HLA-A*68:02					
		HLA-A*02:03					
		HLA-B*07:02					
	TLSLFPIDA	HLA-A*02:01	0.9416	NON-ALLERGEN	NON- TOXIN	Outside	0.16398
		HLA-A*02:03					
	TTHDLHLTP	HLA-A*68:02	1.9824	NON-ALLERGEN	NON- TOXIN	Outside	0.06579
	TTHDLHLTPY	HLA-A*01:01	1.6776	NON-ALLERGEN	NON- TOXIN	Outside	0.06936
		HLA-A*26:01					
		HLA-A*30:02					
		HLA-B*15:01					
		HLA-B*35:01					
		HLA-B*58:01					
		HLA-B*57:01					
		HLA-A*11:01					
		HLA-A*32:01					
		HLA-A*68:01					
		HLA-B*53:01					
	TVLNFLPTH	HLA-A*11:01	0.9166	NON-ALLERGEN	NON- TOXIN	Outside	0.10677
		HLA-A*30:02					
		HLA-A*03:01					
		HLA-A*26:01					
		HLA-A*31:01					
		HLA-A*30:01					
		HLA-A*32:01					
	VAVLDIALSL	HLA-A*02:06	1.2250	NON-ALLERGEN	NON- TOXIN	Outside	0.08425
		HLA-A*02:01					

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Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-B*51:01					
		HLA-A*32:01					
	VLDIALSLA	HLA-A*02:01	1.2505	NON-ALLERGEN	NON- TOXIN	Outside	0.02268
		HLA-A*02:06					
		HLA-A*01:01					
		HLA-A*02:03					
	WYFVEHEPL	HLA-A*23:01	0.6006	NON-ALLERGEN	NON- TOXIN	Outside	0.28551
		HLA-A*24:02					
	WYKTTHDLHL	HLA-A*24:02	0.7072	NON-ALLERGEN	NON- TOXIN	Outside	0.06878
		HLA-A*23:01					
	YILFFPFGLF	HLA-A*26:01	0.7278	NON-ALLERGEN	NON- TOXIN	Outside	0.34972
VP40	YYDELHDLNL	HLA-A*24:02	1.0898	NON-ALLERGEN	NON- TOXIN	Outside	0.13639
		HLA-A*23:01					
		HLA-B*08:01					
		HLA-A*23:01					
	FTYNLTNLVL	HLA-A*68:02	0.8113	NON-ALLERGEN	NON- TOXIN	Outside	0.02796
		HLA-A*24:02					
		HLA-A*23:01					
	GIPAHPLRM	HLA-A*26:01	0.4623	NON-ALLERGEN	NON- TOXIN	Outside	0.07771
	HPAVSIHPNL	HLA-B*07:02	0.9293	NON-ALLERGEN	NON- TOXIN	Outside	0.04005
		HLA-B*53:01					
		HLA-B*35:01					
		HLA-B*51:01					
		HLA-A*68:02					
		HLA-B*08:01					
	HPNLPPIVLP	HLA-B*07:02	0.9180	NON-ALLERGEN	NON- TOXIN	Outside	0.11878
		HLA-B*51:01					
		HLA-B*53:01					
		HLA-B*35:01					
	LAHTVAALL	HLA-B*51:01	0.4095	NON-ALLERGEN	NON- TOXIN	Outside	0.15313
		HLA-B*53:01					
		HLA-A*68:02					
		HLA-B*35:01					
		HLA-B*58:01					
	NPNNGPLLAI	HLA-B*07:02	0.7130	NON-ALLERGEN	NON- TOXIN	Outside	0.01665
		HLA-B*51:01					
		HLA-B*53:01					
		HLA-B*35:01					
	TGIPAHPLR	HLA-A*33:01	0.4004	NON-ALLERGEN	NON- TOXIN	Outside	0.08475
		HLA-A*31:01					
		HLA-A*68:01					
		HLA-A*30:01					
		HLA-A*11:01					
	TPNYVGDLNL	HLA-B*07:02	0.8830	NON-ALLERGEN	NON- TOXIN	Outside	0.07512
		HLA-B*53:01					
		HLA-B*51:01					
		HLA-B*35:01					

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Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
	TQFTHNGQK	HLA-A*30:01	0.6418	NON-ALLERGEN	NON- TOXIN	Inside	0.06339
		HLA-A*11:01					
		HLA-A*03:01					
	TYNLTNLVL	HLA-A*24:02	0.9020	NON-ALLERGEN	NON- TOXIN	Outside	0.03321
		HLA-A*23:01					
		HLA-B*08:01					
	VPAWLPLGI	HLA-B*51:01	1.3515	NON-ALLERGEN	NON- TOXIN	Outside	0.22479
		HLA-B*53:01					
		HLA-B*07:02					
		HLA-B*35:01					
		HLA-A*68:02					
	VPAWLPLGIM	HLA-B*07:02	1.1176	NON-ALLERGEN	NON- TOXIN	Outside	0.30991
		HLA-B*53:01					
		HLA-B*35:01					
		HLA-B*51:01					
VP35	VVLAYANPTL	HLA-A*02:01	0.9123	NON-ALLERGEN	NON- TOXIN	Outside	0.0775
	AKDLALLLF	HLA-B*40:01	0.7224	NON-ALLERGEN	NON- TOXIN	Outside	0.00786
	DLALLLFTH	HLA-A*33:01	0.6685	NON-ALLERGEN	NON- TOXIN	Outside	0.10178
	DLALLLFTHL	HLA-B*51:01	0.7297	NON-ALLERGEN	NON- TOXIN	Outside	0.1418
	DLLIPINRQI	HLA-A*68:02	0.4438	NON-ALLERGEN	NON- TOXIN	Outside	0.21903
	DQVFGANPL	HLA-B*40:01	0.5139	NON-ALLERGEN	NON- TOXIN	Outside	0.18909
	LALLLFTHL	HLA-B*51:01	0.5427	NON-ALLERGEN	NON- TOXIN	Outside	0.1363
		HLA-B*08:01					
		HLA-B*58:01					
		HLA-B*57:01					
	LLIPINRQI	HLA-A*02:03	0.5035	NON-ALLERGEN	NON- TOXIN	Outside	0.13155
		HLA-A*02:01					
		HLA-A*32:01					
		HLA-A*02:06					
		HLA-B*51:01					
		HLA-A*68:02					
		HLA-B*58:01					
		HLA-B*57:01					
	LPGNNTPFHI	HLA-B*51:01	0.4737	NON-ALLERGEN	NON- TOXIN	Outside	0.14325
		HLA-B*53:01					
	NEHGVPPPQ	HLA-B*44:03	0.6767	NON-ALLERGEN	NON- TOXIN	Outside	0.05852
	NNTPFHILA	HLA-A*68:02	0.7753	NON-ALLERGEN	NON- TOXIN	Outside	0.25173
	NTPFHILAQV	HLA-A*68:02	0.4520	NON-ALLERGEN	NON- TOXIN	Outside	0.2302
		HLA-B*51:01					
		HLA-A*02:06					
	PFHILAQVL	HLA-A*24:02	0.4186	NON-ALLERGEN	NON- TOXIN	Outside	0.09681
		HLA-A*23:01					
	RTFDAFLGVV	HLA-A*02:06	0.6155	NON-ALLERGEN	NON- TOXIN	Outside	0.2547
		HLA-A*68:02					
		HLA-A*32:01					
		HLA-A*30:01					
		HLA-A*02:03					

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Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-A*02:01					
		HLA-B*57:01					
VP30	TPFHILAQV	HLA-B*51:01	0.6265	NON-ALLERGEN	NON- TOXIN	Outside	0.15505
		HLA-A*68:02					
		HLA-B*07:02					
		HLA-B*53:01					
		HLA-B*35:01					
		HLA-A*26:01					
		HLA-B*08:01					
	LDETSLRAAL	HLA-B*40:01	0.5565	NON-ALLERGEN	NON- TOXIN	Outside	0.00426
	LTLENLGHI	HLA-A*02:06	1.3198	NON-ALLERGEN	NON- TOXIN	Outside	0.12791
		HLA-A*68:02					
		HLA-B*58:01					
		HLA-B*57:01					
	LTLENLGHIL	HLA-B*57:01	0.8870	NON-ALLERGEN	NON- TOXIN	Outside	0.21701
	LTNRELLLL	HLA-B*58:01	1.1351	NON-ALLERGEN	NON- TOXIN	Outside	0.1212
		HLA-B*57:01					
		HLA-A*32:01					
		HLA-A*01:01					
	LTNRELLLLM	HLA-B*57:01	0.9915	NON-ALLERGEN	NON- TOXIN	Outside	0.1104
		HLA-B*58:01					
	NLTNRELLL	HLA-B*08:01	1.0774	NON-ALLERGEN	NON- TOXIN	Outside	0.1349
		HLA-A*02:01					
		HLA-A*02:03					
	TSLRAALSL	HLA-B*58:01	0.6448	NON-ALLERGEN	NON- TOXIN	Outside	0.01739
		HLA-B*57:01					
		HLA-A*32:01					
VP24	EPFLALRIL	HLA-B*51:01	0.7859	NON-ALLERGEN	NON- TOXIN	Outside	0.17594
		HLA-B*53:01					
		HLA-B*07:02					
		HLA-B*35:01					
		HLA-B*08:01					
	EPFLALRILL	HLA-B*51:01	0.4989	NON-ALLERGEN	NON- TOXIN	Outside	0.2087
		HLA-B*07:02					
		HLA-B*53:01					
	EPSVGGWTV	HLA-B*51:01	0.6941	NON-ALLERGEN	NON- TOXIN	Outside	0.26236
		HLA-B*53:01					
		HLA-B*35:01					
		HLA-B*07:02					
		HLA-A*68:02					
		HLA-B*08:01					
	EPSVGGWTVK	HLA-A*68:01	1.3392	NON-ALLERGEN	NON- TOXIN	Outside	0.31923
	ESVVFGLVA	HLA-A*68:02	1.0973	NON-ALLERGEN	NON- TOXIN	Outside	0.2156
	FLALRILLGV	HLA-A*02:03	0.6712	NON-ALLERGEN	NON- TOXIN	Outside	0.18154
		HLA-A*02:01					
		HLA-A*02:06					
	GWTVKWGNF	HLA-A*24:02	1.7338	NON-ALLERGEN	NON- TOXIN	Outside	0.07747

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Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
		HLA-A*23:01					
	HISPNLLGIY	HLA-A*30:02	1.2031	NON-ALLERGEN	NON- TOXIN	Outside	0.01396
		HLA-A*26:01					
		HLA-A*01:01					
		HLA-B*15:01					
		HLA-B*35:01					
		HLA-A*03:01					
	IEPFLALRI	HLA-B*40:01	0.98	NON-ALLERGEN	NON- TOXIN	Outside	0.16111
		HLA-B*44:03					
		HLA-B*44:02					
	IEPFLALRIL	HLA-B*40:01	0.704	NON-ALLERGEN	NON- TOXIN	Outside	0.2525
		HLA-B*51:01					
		HLA-B*07:02					
	ISPNLLGIY	HLA-A*30:02	1.0528	NON-ALLERGEN	NON- TOXIN	Outside	0.07501
		HLA-A*01:01					
		HLA-A*26:01					
		HLA-B*15:01					
		HLA-B*58:01					
		HLA-B*57:01					
	ISPNLLGIYL	HLA-B*07:02	0.9699	NON-ALLERGEN	NON- TOXIN	Outside	0.11035
		HLA-B*53:01					
		HLA-B*51:01					
	KEPSVGGWTV	HLA-B*40:01	0.9418	NON-ALLERGEN	NON- TOXIN	Outside	0.14465
		HLA-B*07:02					
	LALRILLGV	HLA-A*02:06	0.6567	NON-ALLERGEN	NON- TOXIN	Outside	0.17808
		HLA-B*51:01					
	LQQSLIPGFR	HLA-A*31:01	0.6741	NON-ALLERGEN	NON- TOXIN	Outside	0.00129
	LRILLGVALK	HLA-A*03:01	0.8469	NON-ALLERGEN	NON- TOXIN	Outside	0.11964
	LSESVVFGL	HLA-B*58:01	0.7713	NON-ALLERGEN	NON- TOXIN	Outside	0.06369
	PFLALRILL	HLA-A*23:01	0.7775	NON-ALLERGEN	NON- TOXIN	Outside	0.17953
		HLA-A*24:02					
	PSVGGWTVKW	HLA-B*58:01	1.0721	NON-ALLERGEN	NON- TOXIN	Outside	0.24158
		HLA-B*57:01					
	QSLIPGFRSI	HLA-A*02:03	0.7881	NON-ALLERGEN	NON- TOXIN	Outside	0.20974
		HLA-B*57:01					
		HLA-B*58:01					
		HLA-A*32:01					
		HLA-A*02:06					
		HLA-A*68:02					
		HLA-A*02:01					
	RILLGVALK	HLA-A*03:01	0.8602	NON-ALLERGEN	NON- TOXIN	Outside	0.08364
		HLA-A*30:01					
		HLA-A*11:01					
		HLA-A*31:01					
	SESVVFGLV	HLA-B*40:01	0.9092	NON-ALLERGEN	NON- TOXIN	Outside	0.16036
		HLA-B*44:02					
		HLA-B*44:03					

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Protein Name	Peptide	Allele	Antigenicity	Allergenicity	Toxicity	Topology	Immunogenicity
	SLIPGFRSI	HLA-A*02:03	0.9273	NON-ALLERGEN	NON- TOXIN	Outside	0.12226
		HLA-A*32:01					
		HLA-A*02:06					
		HLA-A*02:01					
		HLA-B*08:01					
		HLA-A*68:02					
		HLA-B*15:01					
		HLA-A*26:01					
		HLA-A*30:01					
		HLA-A*23:01					
		HLA-B*51:01					
		HLA-A*24:02					
	SLIPGFRSIV	HLA-A*02:03	0.4106	NON-ALLERGEN	NON- TOXIN	Outside	0.1659
		HLA-A*02:06					
		HLA-A*02:01					
		HLA-A*68:02					
	SPNLLGIYL	HLA-B*07:02	0.8416	NON-ALLERGEN	NON- TOXIN	Outside	0.118
		HLA-B*53:01					
		HLA-B*35:01					
		HLA-B*51:01					
		HLA-B*08:01					
	SPNLLGIYLT	HLA-B*07:02	1.0929	NON-ALLERGEN	NON- TOXIN	Outside	0.12462
	SVVFGLVAEA	HLA-A*68:02	0.9088	NON-ALLERGEN	NON- TOXIN	Outside	0.28378
		HLA-A*02:06					
		HLA-A*02:03					
	TGMTLLHHL	HLA-B*08:01	0.5236	NON-ALLERGEN	NON- TOXIN	Outside	0.00702
		HLA-B*51:01					
		HLA-A*68:02					
		HLA-A*02:06					
	VLSESVVFGL	HLA-A*02:01	0.6037	NON-ALLERGEN	NON- TOXIN	Outside	0.08361
		HLA-A*02:03					
		HLA-A*02:06					
	VVFGLVAEA	HLA-A*02:06	0.9777	NON-ALLERGEN	NON- TOXIN	Outside	0.19168
		HLA-A*68:02					
		HLA-A*02:03					
		HLA-A*02:01					
	VVFGLVAEAV	HLA-A*68:02	0.8754	NON-ALLERGEN	NON- TOXIN	Outside	0.24569
		HLA-A*02:06					
		HLA-A*02:03					
		HLA-A*02:01					
	WLLLEVTSA	HLA-A*02:01	1.1675	NON-ALLERGEN	NON- TOXIN	Outside	0.0577
		HLA-A*02:03					
		HLA-A*02:06					
	WTVKWGNFVF	HLA-B*57:01	1.7660	NON-ALLERGEN	NON- TOXIN	Outside	0.16193
		HLA-B*58:01					

Table S3. Predicted 36 Anticipated HTL candidates.

Protein Name	Peptide	Allele	Antigenecity	Allergenecity	Toxicity	Topology	IFN-Ɣ	IL-4	IL-10
GP	LEIASNNQPQNVDSV	HLA-DRB3*02:02	0.7993	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	LELLLRVTTEERTFS	HLA-DRB5*01:01	0.7135	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	QTAKSLELLLRVTTE	HLA-DPA1*02:01/DPB1*05:01	0.7143	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	TAKSLELLLRVTTEE	HLA-DRB1*11:01	0.7734	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
NP	FILKFVLIHQGVNLV	HLA-DRB1*15:01	0.6203	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	FLEVIPNEPHYSPLI	HLA-DRB3*02:02	0.5692	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*13:02							
L	AEKLEYLAPSYRNFS	HLA-DRB5*01:01	1.1197	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*02:01/DPB1*05:01							
		HLA-DRB1*12:01							
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*01:03/DPB1*04:01							
	APSYRNFSFSLKEKE	HLA-DPA1*02:01/DPB1*05:01	2.2255	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*02:01/DPB1*14:01							
		HLA-DPA1*02:01/DPB1*01:01							
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*01:03/DPB1*04:01							
		HLA-DPA1*03:01/DPB1*04:02							
	EFFLTLGFPSLESSF	HLA-DPA1*01:03/DPB1*04:01	0.9614	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*02:01/DPB1*01:01							
		HLA-DPA1*03:01/DPB1*04:02							
	EKLEYLAPSYRNFSF	HLA-DRB5*01:01	1.6017	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*01:03/DPB1*04:01							
	FAEKLEYLAPSYRNF	HLA-DRB5*01:01	1.0187	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*02:01/DPB1*05:01							
		HLA-DRB1*12:01							
	FQILIKSLDAHEDFD	HLA-DQA1*03:01/DQB1*03:02	0.6609	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*04:05							
		HLA-DQA1*01:01/DQB1*05:01							
		HLA-DQA1*04:01/DQB1*04:02							
		HLA-DRB4*01:01							
		HLA-DQA1*05:01/DQB1*02:01							

Continued

Protein Name	Peptide	Allele	Antigenecity	Allergenecity	Toxicity	Topology	IFN-Ɣ	IL-4	IL-10
		HLA-DRB1*12:01							
		HLA-DPA1*02:01/DPB1*01:01							
		HLA-DRB1*04:01							
		HLA-DRB1*01:01							
	FSFSLKEKELNIGRT	HLA-DRB1*11:01	2.3360	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*08:02							
		HLA-DPA1*01:03/DPB1*02:01							
	GSLASIGTSFERGTS	HLA-DRB1*04:05	0.7559	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DQA1*03:01/DQB1*03:02							
	HEFFLTLGFPSLESS	HLA-DPA1*01:03/DPB1*04:01	0.8885	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*02:01/DPB1*01:01							
		HLA-DPA1*03:01/DPB1*04:02							
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*02:01/DPB1*05:01							
	KLEYLAPSYRNFSFS	HLA-DPA1*01:03/DPB1*02:01	1.809	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*01:03/DPB1*04:01							
	NWKFIGTDLELFGIA	HLA-DPA1*02:01/DPB1*01:01	1.1489	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*01:03/DPB1*04:01							
		HLA-DPA1*03:01/DPB1*04:02							
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*02:01/DPB1*05:01							
		HLA-DQA1*03:01/DQB1*03:02							
		HLA-DQA1*05:01/DQB1*02:01							
	PSYRNFSFSLKEKEL	HLA-DPA1*02:01/DPB1*14:01	2.2592	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*02:01/DPB1*05:01							
		HLA-DPA1*02:01/DPB1*01:01							
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*01:03/DPB1*04:01							
	QGSLASIGTSFERGT	HLA-DRB1*04:05	0.7819	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DQA1*03:01/DQB1*03:02							
		HLA-DRB1*04:01							
		HLA-DQA1*05:01/DQB1*02:01							
	SGFIYFGKKQYLNGV	HLA-DRB1*13:02	0.7480	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER

Protein Name	Peptide	Allele	Antigenecity	Allergenecity	Toxicity	Topology	IFN-Ɣ	IL-4	IL-10
		HLA-DRB1*08:02							
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*02:01/DPB1*05:01							
		HLA-DRB1*11:01							
		HLA-DRB1*12:01							
VP40	AFTLEAIIDISAYNE	HLA-DQA1*01:01/DQB1*05:01	0.5822	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DQA1*03:01/DQB1*03:02							
		HLA-DRB1*04:01							
		HLA-DRB1*04:05							
	NFEYPLAHTVAALLT	HLA-DRB1*07:01	0.5362	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DPA1*02:01/DPB1*14:01							
		HLA-DRB1*09:01							
		HLA-DPA1*01:03/DPB1*04:01							
	SNFEYPLAHTVAALL	HLA-DPA1*02:01/DPB1*14:01	0.4400	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*07:01							
		HLA-DPA1*02:01/DPB1*01:01							
		HLA-DPA1*01:03/DPB1*04:01							
		HLA-DPA1*01:03/DPB1*02:01							
		HLA-DPA1*03:01/DPB1*04:02							
		HLA-DRB1*09:01							
		HLA-DPA1*02:01/DPB1*05:01							
VP35	EITPVLKMGRTLEAI	HLA-DRB1*11:01	0.9411	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	LHEITPVLKMGRTLE	HLA-DRB1*13:02	0.6485	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*03:01							
		HLA-DRB1*11:01							
	SEMLAKYDHLVISTG	HLA-DRB1*15:01	0.6463	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
VP30	ASYDHFILPQSQGKG	HLA-DQA1*04:01/DQB1*04:02	0.6038	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DQA1*01:02/DQB1*06:02							
	ISVQASYDHFILPQS	HLA-DQA1*01:01/DQB1*05:01	0.5821	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	LTNRELLLLMARKML	HLA-DRB1*12:01	0.8097	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*01:01							
	QASYDHFILPQSQGK	HLA-DQA1*04:01/DQB1*04:02	0.5531	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	SSISVQASYDHFILP	HLA-DQA1*01:01/DQB1*05:01	0.7246	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*09:01							

Continued

Protein Name	Peptide	Allele	Antigenecity	Allergenecity	Toxicity	Topology	IFN-Ɣ	IL-4	IL-10
		HLA-DRB1*07:01							
	VQASYDHFILPQSQG	HLA-DQA1*04:01/DQB1*04:02	0.5242	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
VP24	EVTSAIHISPNLLGI	HLA-DRB1*04:01	1.2318	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB4*01:01							
		HLA-DRB1*13:02							
		HLA-DRB3*02:02							
		HLA-DRB1*15:01							
		HLA-DRB1*04:05							
		HLA-DRB1*08:02							
		HLA-DRB1*03:01							
		HLA-DRB1*09:01							
		HLA-DRB1*07:01							
		HLA-DRB1*11:01							
	PSVGGWTVKWGNFVF	HLA-DQA1*04:01/DQB1*04:02	1.4546	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
	SAIHISPNLLGIYLT	HLA-DRB1*04:01	1.2424	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB4*01:01							
		HLA-DRB1*15:01							
		HLA-DRB1*13:02							
		HLA-DRB3*02:02							
		HLA-DRB1*04:05							
		HLA-DRB1*03:01							
	TSAIHISPNLLGIYL	HLA-DRB1*04:01	1.0230	NON -ALLERGEN	NON-TOXIN	OUTSIDE	INDUCER	INDUCER	INDUCER
		HLA-DRB1*13:02							
		HLA-DRB1*15:01							
		HLA-DRB4*01:01							
		HLA-DRB3*02:02							
		HLA-DRB1*08:02							
		HLA-DRB1*04:05							
		HLA-DRB1*03:01							
		HLA-DRB1*11:01							
		HLA-DRB1*09:01							
		HLA-DRB5*01:01							
		HLA-DRB1*07:01							

Table S4. Predicted 33 potential B-cell epitopes.

Protein Name	Start	End	Peptide	Length	Antigenicity	Allergenicity	TMHMM	Toxicity
VP40	77	87	VKGVPAWLPLG	11	1.0860	NON-ALLERGEN	Outside	NON-TOXIN
	131	136	PAHPLR	6	0.9093	NON-ALLERGEN	Outside	NON-TOXIN
VP35	211	220	KDLALLLFTH	10	0.7232	NON-ALLERGEN	Outside	NON-TOXIN
	131	137	YDHLVIS	7	0.9748	NON-ALLERGEN	Outside	NON-TOXIN
	312	319	GWVCVYSS	8	0.7419	NON-ALLERGEN	Outside	NON-TOXIN
	44	53	GLQCSPCLMS	10	0.7797	NON-ALLERGEN	Outside	NON-TOXIN
VP30	68	74	SKPPPPP	7	1.3163	NON-ALLERGEN	Outside	NON-TOXIN
	90	103	PTCNRDHDLDNLTN	14	0.4668	NON-ALLERGEN	Outside	NON-TOXIN
	68	73	SKPPPP	6	0.7310	NON-ALLERGEN	Outside	NON-TOXIN
VP 24	172	180	DHGKPSSIE	9	0.7245	NON-ALLERGEN	Outside	NON-TOXIN
	172	177	DHGKPS	6	0.7093	NON-ALLERGEN	Outside	NON-TOXIN
	495	501	FPDKNGD	7	0.732	NON-ALLERGEN	Outside	NON-TOXIN
GLY	55	65	SGQKVADSPLE	11	0.4541	NON-ALLERGEN	Outside	NON-TOXIN
	211	217	CFGILQE	7	1.4569	NON-ALLERGEN	Outside	NON-TOXIN
	445	451	FPFLDGL	7	0.7907	NON-ALLERGEN	Outside	NON-TOXIN
	527	537	AAGLSWIPFFG	11	0.7192	NON-ALLERGEN	Outside	NON-TOXIN
	569	576	SLELLLRV	8	0.725	NON-ALLERGEN	Outside	NON-TOXIN
	241	247	DSGVALH	7	0.6934	NON-ALLERGEN	Outside	NON-TOXIN
NP	286	293	GINNLEHG	8	1.352	NON-ALLERGEN	Outside	NON-TOXIN
	109	118	YSPLILALKT	10	1.3161	NON-ALLERGEN	Outside	NON-TOXIN
	243	253	GVALHPLVRTS	11	0.6627	NON-ALLERGEN	Outside	NON-TOXIN
	519	528	FLRYQELPPV	10	1.5382	NON-ALLERGEN	Outside	NON-TOXIN
	555	564	VDLPPPPLYA	10	0.7098	NON-ALLERGEN	Outside	NON-TOXIN
L	138	145	FSPKENPY	8	1.578	NON-ALLERGEN	Outside	NON-TOXIN
	506	519	LGYNPPVRFQSKRV	14	0.5897	NON-ALLERGEN	Outside	NON-TOXIN
	148	158	GMLLLVHLSQL	11	0.5456	NON-ALLERGEN	Outside	NON-TOXIN
	93	105	DFKYLLPSELVKY	13	0.545	NON-ALLERGEN	Outside	NON-TOXIN
	902	915	KLIALITPQVLGGL	14	0.5404	NON-ALLERGEN	Outside	NON-TOXIN
	1180	1187	EGSNDTEL	8	0.5079	NON-ALLERGEN	Outside	NON-TOXIN
	1516	1522	LTFLISS	7	0.4477	NON-ALLERGEN	Outside	NON-TOXIN
	262	273	LEDSDGADYPPL	12	0.4357	NON-ALLERGEN	Outside	NON-TOXIN
	1180	1186	EGSNDTE	7	0.4264	NON-ALLERGEN	Outside	NON-TOXIN
	478	484	ISDLSIF	7	0.404	NON-ALLERGEN	Outside	NON-TOXIN

Table S5. B cell conformation epitopes.

No.	Residues	Number of residues	Score
1	A:F232, A:Q236, A:E237	3	0.776
2	A:V13, A:G15, A:R234, A:Y235, A:L238, A:P239, A:P240,A:V241,A:K242, A:G244, A:I245, A:N246, A:N247, A:L248, A:H250, A:G251	16	0.743
3	A:G200, A:P201, A:G202, A:P203, A:G204, A:A205, A:F206	7	0.639
4	A:G16,A:R17,A:S22,A:L24,A:P25,A:K26,A:E28,A:Q29,A:I30,A:G31,A:K32, A:C33, A:S34, A:T35, A:G37, A:R38, A:C41, A:R42, A:K44, A:K45, A:A46, A:A47,A:Y48,A:A49, A:P50, A:I51, A:D52, A:F53, A:D54, A:P55, A:V56, A:P57, A:A58, A:A59, A:Y60, A:A62, A:A63, A:S66,A:H74,A:P77,A:L80,A:A81,A:A84,A:A85,A:Y86,A:T87,A:T88,A:H89,A:D90,A:L91,A:H92,A:L93,A:T94,A:P95,A:A96,A:A97,A:Y98,A:G99,A:L113,A:E114,A:N115,A:G117,A:H118,A:I119,A:G120,A:P121,A:G122,A:P123,A:G124,A:L125,A:E126,A:I127,A:A128,A:S129,A:N130,A:N131,A:Q132,A:P133,A:Q134,A:N135,A:V136,A:D137,A:S138,A:V139,A:G140,A:P141,A:G142,A:P143,A:G144,A:F145,A:I152,A:H153,A:Q154,A:G155,A:V156,A:N157,A:L158, A:V159,A:G160,A:P161,A:G162,A:P163,A:G164,A:A165,A:P166,A:S167,A:Y168,A:R169,A:K178, A:E179, A:G180, A:P181, A:G182, A:P183,A:G184,A:E185, A:I186, A:M192, A:N218, A:E219, A:K220, A:K221	122	0.638
5	A:G1, A:I2, A:I3	3	0.619

The constructed mRNA vaccine of Marburg Virus

GCUUCUGCUGGGGGACAGAGUGAGUGGCCUGGCCACCAUGGAUGGACGCCAUGAAGAGAGGCCUGU-
GUUGUGCUCCUGUGUGGCUGUCUUUGUGUCUCCCAGUGGCAUCAUCAACACCCUGCAGAAGUACUA-
CUGCCGCGUGCGCGGCGGCCGCUGCGCCGUGCUGAGCUGCCUGCCCAAGGAGGAGCAGAUCGGCAAGUG-
CAGCACCCGCGGCCGCAAGUGCUGCCGCCGCAAGAAGGCCGCCUACGCCCCCAUCGACUUCGACCCCGUGC-
CCGCCGCCUACCUGGCCGCCGGCCUGAGCUGGAUCCCCUUCGCCGCCUACCACUACAGCCCCCUGAUCCUG-
GCCCUGAAGGCCGCCUACACCACCCACGACCUGCACCUGACCCCCGCCGCCUACGGCUGGACCGUGAAGUG-
GGGCAACUUCGCCGCCUACCUGACCCUGGAGAACCUGGGCCACAUCGGCCCCGGCCCCGGCCUGGAGAUC-
GCCAGCAACAACCAGCCCCAGAACGUGGACAGCGUGGGCCCCGGCCCCGGCUUCAUCCUGAAGUUC-
GUGCUGAUCCACCAGGGCGUGAACCUGGUGGGCCCCGGCCCCGGCGCCCCCAGCUACCGCAACUUCA-
GCUUCAGCCUGAAGGAGAAGGAGGGCCCCGGCCCCGGCGAGAUCACCCCCGUGCUGAAGAUGGGCCG-
CACCCUGGAGGCCAUCGGCCCCGGCCCCGGCGCCUUCACCCUGGAGGCCAUCAUCGACAUCAGCGCCUA-
CAACGAGAAGAAGAUCUGCUUCGGCAUCCUGCAGGAGAAGAAGUUCCUGCGCUACCAGGAGCUGC-
CCCCCGUGAAGAAGGGCAUCAACAACCUGGAGCACGGCGAAAGGUGAGGCCAGGCGGGAAAGCUGGAGU-
GAGGGAGGCGGAGGAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA